

KMS STATE UNIQUENESS AND PHASE TRANSITIONS IN LOCALIZED BOST–CONNES SYSTEMS: FROM CHEBOTAREV DENSITY TO THE HARDY–LITTLEWOOD CONJECTURE

RUQING CHEN

ABSTRACT. For a number field K and an arbitrary set S of finite primes of K we study the S -local Bost–Connes system $\mathcal{A}_{K,S}$, whose partition function is the partial Dedekind product $\zeta_{K,S}(\beta) = \prod_{\mathfrak{p} \in S} (1 - \mathrm{N}\mathfrak{p}^{-\beta})^{-1}$ and whose symmetry group is $G_S = \mathbb{I}_S / \overline{K_{S,+}^\times}$, an extension of the subgroup Cl_S^+ of the narrow class group generated by S by $U_S / \overline{\mathcal{O}_{K,+}^\times}$; for S the set of all primes, $G_S \cong \mathrm{Gal}(K^{ab}/K)$. We classify the KMS_β states completely and unconditionally, for every K , every S and every $\beta > 0$: the simplex is affinely isomorphic to $\mathrm{Prob}(G_S / \Xi_\beta^+)$, where

$$\Xi_\beta(K, S) = \left\{ \chi \in \widehat{G_S} : \sum_{\mathfrak{p} \in S} \mathrm{N}\mathfrak{p}^{-\beta} (1 - \mathrm{Re} \chi(\mathrm{Frob}_{\mathfrak{p}})) < \infty \right\}.$$

Uniqueness therefore holds if and only if, for every nontrivial finite abelian extension L/K unramified outside $S \cup \infty$, the primes of S that do not split completely in L have divergent β -weighted sum; for S the set of all primes this is the Chebotarev density theorem, and at $\beta = 1$ it is the non-vanishing $L(1, \chi) \neq 0$. The criterion refutes the natural expectation that uniqueness should follow from the divergence $\zeta_{K,S}(\beta) = \infty$: we exhibit sets S with $\zeta_{K,S}(\beta) = \infty$ and with $\sigma(I_S)$ dense in G_S for which the KMS_β simplex is a segment at every $\beta > 0$, the two extreme points being separated by an explicit pointwise invariant whose almost-everywhere definedness is a Borel–Cantelli condition. Two mechanisms are isolated: over a field with $h_K^+ > 1$ the Hilbert class field supplies obstructing characters that are everywhere unramified, a phenomenon impossible over $\mathbb{Q}$; and over $\mathbb{Q}$ the separation between topological density and measure-theoretic ergodicity is governed by a Kakutani dichotomy. We show that $\beta \mapsto \Xi_\beta$ is non-decreasing and construct systems with any prescribed finite number of phase transitions. Finally we treat sparse sets: for Brun-summable S with critical exponent 1 the partition function is finite at the critical point, so the low-temperature phase is closed—the opposite of the classical Bost–Connes system—and for $S = \{2\} \cup \mathcal{SG}$, $\mathcal{SG}$ the Sophie Germain primes, the existence and location of the transition are equivalent to a Hardy–Littlewood type density statement.

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1. INTRODUCTION

The Bost–Connes system [1] is a C^* -dynamical system $(\mathcal{A}_{BC}, \sigma)$ with partition function $\zeta(\beta)$ and symmetry group $\widehat{\mathbb{Z}}^\times \cong \text{Gal}(\mathbb{Q}^{ab}/\mathbb{Q})$, exhibiting a phase transition at $\beta = 1$: for $0 < \beta \leq 1$ the KMS_β state is unique and of type III_1 ; for $\beta > 1$ the extremal KMS_β states are of type I_∞ and are permuted freely and transitively by $\widehat{\mathbb{Z}}^\times$. Number field analogues were constructed by Ha–Paugam [7] and analysed by Laca–Larsen–Neshveyev [11], with ζ_K in place of ζ and $\text{Gal}(K^{ab}/K)$ in place of $\text{Gal}(\mathbb{Q}^{ab}/\mathbb{Q})$.

This paper studies the systems obtained by replacing the set of all primes by an arbitrary subset S . Two questions organize the analysis. First, when is the KMS_β state unique? Second, what does the answer depend on?

The received expectation is that uniqueness should be the shadow of the divergence $\zeta_{K,S}(\beta) = \infty$: when the partition function diverges there are no Gibbs states, every scaling-invariant measure gives the units measure zero, and one expects the resulting measure to be forced. We show that this expectation is false. The correct answer is Theorem 3.6: the KMS_β simplex is $\text{Prob}(G_S/\Xi_\beta^\perp)$ with Ξ_β as in (1.1) below, so that uniqueness is equivalent to the vanishing of an obstruction group defined by a convergence condition on a Dirichlet series indexed by S . Divergence of $\zeta_{K,S}(\beta)$ is not enough; what is needed is that, for each nontrivial character, the primes of S detecting it are themselves β -abundant. For S the set of all primes this is supplied by the Chebotarev density theorem, and the classical uniqueness statement is recovered; for sparse S it can fail, and we exhibit the failure explicitly.

Main results. Let K be a number field, S a set of finite primes, G_S the S -local symmetry group of Definition 2.1, and $\sigma : I_S \rightarrow G_S$ the canonical homomorphism from the group of fractional ideals supported on S (Lemma 2.10). Set

$$(1.1) \quad \Xi_\beta(K, S) = \left\{ \chi \in \widehat{G_S} : \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}})|^2 < \infty \right\}.$$

Since $|\chi(\sigma_{\mathfrak{p}})| = 1$, the summand may be written either way:

$$(1.2) \quad |1 - \chi(\sigma_{\mathfrak{p}})|^2 = 2(1 - \text{Re } \chi(\sigma_{\mathfrak{p}})) = 2(1 - \text{Re } \chi(\text{Frob}_{\mathfrak{p}})),$$

the last step because $\chi(\sigma_{\mathfrak{p}}) = \chi(\text{Frob}_{\mathfrak{p}})^{-1} = \overline{\chi(\text{Frob}_{\mathfrak{p}})}$ by Lemma 3.8. The factor 2 and the inversion do not affect convergence, so the form used in the abstract and (1.1) define the same subgroup.

- **Classification** (Theorem 3.6). For every K, S and $\beta > 0$, the KMS_β simplex of $\mathcal{A}_{K,S}$ is affinely isomorphic to $\text{Prob}(G_S/\Xi_\beta(K, S)^\perp)$. Uniqueness holds iff $\Xi_\beta = \{1\}$; if $\zeta_{K,S}(\beta) <$

- ∞ then $\Xi_\beta = \widehat{G_S}$ and one recovers the Gibbs states of type I_∞ permuted freely and transitively by G_S . The two classical regimes are the two ends of a single formula.
- **Chebotarev criterion** (Theorem 3.9, Corollary 3.10). Uniqueness at β is equivalent to (3.3): for every nontrivial finite abelian L/K unramified outside $S \cup \infty$, the primes of S not splitting completely in L have divergent β -weighted sum. For $S = P_K$ and $\beta \leq 1$ this follows from Chebotarev, and at $\beta = 1$ it is exactly $L(1, \chi) \neq 0$ (Remark 3.11).
 - **Unramified class-group obstruction** (§4.1). If $h_K^+ > 1$, characters pulled back from the narrow class group are unramified everywhere and obstruct uniqueness whenever the primes of S concentrate β -negligibly in a proper subgroup. Over $\mathbb{Q}$ this cannot happen: every obstruction requires a modulus supported on S , hence ramification. An explicit example over $\mathbb{Q}(\sqrt{-5})$ is Example 4.2.
 - **Topology versus measure** (§4.2). Density of $\sigma(I_S)$ in G_S is necessary but not sufficient for uniqueness. Over $\mathbb{Q}$ we construct $S = \{3\} \cup P_1 \cup R$ with $\sigma(I_S)$ dense, compute $\Xi_\beta = \{1, \chi\}$ exactly, and exhibit the pointwise invariant Ψ of (4.1); its almost-everywhere definedness is a Borel–Cantelli condition, while density needs only $R \neq \emptyset$. The underlying dichotomy is Kakutani’s (Remark 4.15). We also show $\beta \mapsto \Xi_\beta$ is non-decreasing and construct cascades of transitions (Proposition 4.16).
 - **Sparse sets** (§5). The critical inverse temperature is the convergence exponent $\beta_c(S)$. If $\sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta_c} < \infty$ —which holds for Brun-summable S with $\beta_c = 1$ —then the type I_∞ phase is the *closed* half-line $[\beta_c, \infty)$ and the partition function does not diverge at the transition, in contrast with Bost–Connes, where $\beta_c = 1$ lies in the type III phase (Remark 5.3). For $S_\infty = \{2\} \cup \mathcal{SG}$ the Brun–Halberstam–Richert bound gives $\beta_c \leq 1$ unconditionally, a lower bound of Hardy–Littlewood strength gives $\beta_c = 1$, and the existence of a transition is equivalent to $\mathcal{SG}$ having positive convergence exponent (Theorem 5.5). We stress in Remark 5.7 that this is an encoding, not a method.

Conventions. The Artin map is normalized *arithmetically*: for a finite abelian L/K and a prime $\mathfrak{p}$ unramified in L , a uniformizer at $\mathfrak{p}$ (as an idele trivial elsewhere) is sent to the arithmetic Frobenius $\text{Frob}_\mathfrak{p}$, the element of $\text{Gal}(L/K)$ inducing $x \mapsto x^{N\mathfrak{p}}$ on the residue field. All Galois groups are of abelian extensions inside a fixed K^{ab} , so $\text{Frob}_\mathfrak{p}$ is well defined. Sets of primes are always sets of finite primes. N is the absolute norm. Characters of compact abelian groups are continuous. The group G_S is compact (Proposition 2.5) and totally disconnected— $\mathbb{I}_S$ is a totally disconnected locally compact group, so by van Dantzig’s theorem it has a basis of compact open subgroups at the identity, whose images form such a basis in the Hausdorff quotient G_S —hence profinite; so every character of G_S has finite order (Lemma 3.8).

Notation for primes is fixed as follows and used without further comment. In the general number-field setting, primes of K are written $\mathfrak{p}, \mathfrak{q} \in S \subseteq P_K$ with absolute norm $N\mathfrak{p}$. When a statement is specialized to $K = \mathbb{Q}$ —that is, throughout §4.2 and §5—rational primes are written ℓ and $N\ell = \ell$, except that a Sophie Germain prime is written p , so that the associated safe prime is $2p + 1$. A fraktur $\mathfrak{p}$ appearing inside a specialized section is always a back-reference to the general statement being invoked.

Throughout, S is countable, so all the measure spaces below are standard Borel and disintegrations exist.

2. THE S -LOCAL SYSTEM OVER A NUMBER FIELD

2.1. The symmetry group. Let K be a number field of degree d over $\mathbb{Q}$, $\mathcal{O}_K$ its ring of integers, P_K the set of nonzero prime ideals. Fix $\emptyset \neq S \subseteq P_K$ and write:

- $\widehat{\mathcal{O}}_S = \prod_{\mathfrak{p} \in S} \mathcal{O}_\mathfrak{p}$, $U_S = \prod_{\mathfrak{p} \in S} \mathcal{O}_\mathfrak{p}^\times$ (compact);

- I_S = the group of fractional ideals supported on S , free abelian on S ; $J_S \subset I_S$ the monoid of integral ones;
- $\mathbb{I}_S = \prod'_{\mathfrak{p} \in S} K_{\mathfrak{p}}^{\times}$ (restricted product with respect to the $\mathcal{O}_{\mathfrak{p}}^{\times}$), with $\text{div} : \mathbb{I}_S \twoheadrightarrow I_S$ and $\ker = U_S$;
- $K_S^{\times} = \{\alpha \in K^{\times} : v_{\mathfrak{p}}(\alpha) = 0 \ \forall \mathfrak{p} \notin S\}$, and $K_{S,+}^{\times}$ its subgroup of totally positive elements, embedded diagonally in $\mathbb{I}_S$.

Definition 2.1. The S -local symmetry group is the compact abelian group

$$G_S := \mathbb{I}_S / \overline{K_{S,+}^{\times}},$$

with quotient map $r : \mathbb{I}_S \rightarrow G_S$.

Remark 2.2. The closure in Definition 2.1 is not cosmetic. The diagonal image of $K_{S,+}^{\times}$ in $\mathbb{I}_S$ is in general not closed: its intersection with U_S has closure $\overline{\mathcal{O}_{K,+}^{\times}}$, the closure of the totally positive units, which by Proposition 4.17 can be a $\mathbb{Z}_p$ -module of positive rank. Quotienting by $K_{S,+}^{\times}$ itself would leave a non-Hausdorff quotient, and Proposition 2.5 would fail. Two cases where the distinction disappears: $K = \mathbb{Q}$, where $\mathcal{O}_{\mathbb{Q},+}^{\times} = \{1\}$ and $\mathbb{I}_S = \widehat{\mathbb{Z}}_S^{\times} \times \mathbb{Q}_{>0}^S$ splits, so that $K_{S,+}^{\times}$ is already closed; and K imaginary quadratic, where $\mathcal{O}_K^{\times}$ is finite. In both, G_S is computed without any limiting process, which is one reason the phenomena of §4.3 are invisible over $\mathbb{Q}$.

The choice of $K_{S,+}^{\times}$ rather than $K_S^{\times}$ is not a convention: it is forced by the requirement that for $S = P_K$ one recovers the symmetry group of [7, 11].

Lemma 2.3. $\mathbb{A}_{K,f}^{\times} / \overline{K_+^{\times}} \cong \text{Gal}(K^{ab}/K)$, the isomorphism being induced by the Artin map. By contrast $\mathbb{A}_{K,f}^{\times} / \overline{K^{\times}}$ is the quotient of $\text{Gal}(K^{ab}/K)$ by the image of $\{\pm 1\}^{r_1}$, a proper quotient whenever K has a real place; for $K = \mathbb{Q}$ it equals $\widehat{\mathbb{Z}}^{\times} / \{\pm 1\}$.

Proof. By class field theory $\text{Gal}(K^{ab}/K) = \mathbb{A}_K^{\times} / \overline{K^{\times} K_{\infty}^{\times, \circ}}$, where $K_{\infty}^{\times} = \prod_{v|\infty} K_v^{\times}$ and $K_{\infty}^{\times, \circ}$ is its identity component. Consider the composite $j : \mathbb{A}_{K,f}^{\times} \hookrightarrow \mathbb{A}_K^{\times} \rightarrow \text{Gal}(K^{ab}/K)$.

j is surjective. Since $K_{\infty}^{\times} / K_{\infty}^{\times, \circ} \cong \{\pm 1\}^{r_1}$, every class is represented by some (a_f, ε) with $\varepsilon \in \{\pm 1\}^{r_1}$. By weak approximation choose $\alpha \in K^{\times}$ whose sign vector at the real places is ε . Then $(a_f, \varepsilon) \sim (\alpha^{-1} a_f, \alpha_{\infty}^{-1} \varepsilon)$ and $\alpha_{\infty}^{-1} \varepsilon$ is totally positive, hence lies in $K_{\infty}^{\times, \circ}$; so the class equals that of $(\alpha^{-1} a_f, 1)$, which is in the image of j .

$\ker j = \overline{K_+^{\times}}$. If $(a_f, 1) \in \overline{K^{\times} K_{\infty}^{\times, \circ}}$ then, comparing archimedean components, a_f lies in the closure of the set of $\alpha \in K^{\times}$ with $\alpha_{\infty} \in K_{\infty}^{\times, \circ}$, i.e. with α totally positive. The converse inclusion is clear.

For the second statement, $\mathbb{A}_{K,f}^{\times} / \overline{K^{\times}}$ is the further quotient of $\mathbb{A}_{K,f}^{\times} / \overline{K_+^{\times}}$ by the image of $K^{\times}$, which is generated by classes of elements of every sign vector; for $K = \mathbb{Q}$, $\mathbb{A}_f^{\times} = \widehat{\mathbb{Z}}^{\times} \times \mathbb{Q}_{>0}$ and $\mathbb{Q}^{\times} = \{\pm 1\} \times \mathbb{Q}_{>0}$, giving $\widehat{\mathbb{Z}}^{\times} / \{\pm 1\}$. $\square$

Corollary 2.4. For $S = P_K$ one has $\mathbb{I}_{P_K} = \mathbb{A}_{K,f}^{\times}$ and $G_{P_K} \cong \text{Gal}(K^{ab}/K)$. For $K = \mathbb{Q}$ and any S , $\mathbb{I}_S = \widehat{\mathbb{Z}}_S^{\times} \times \mathbb{Q}_{>0}^S$ uniquely, so $G_S = \widehat{\mathbb{Z}}_S^{\times}$.

Proposition 2.5. Let $\text{Cl}_S^+ \leq \text{Cl}_K^+$ be the subgroup of the narrow class group generated by the classes of the primes of S , and $\mathcal{O}_{K,+}^{\times}$ the group of totally positive units. Then $K_{S,+}^{\times} \cap U_S = \mathcal{O}_{K,+}^{\times}$ and there is an exact sequence of compact abelian groups

$$(2.1) \quad 1 \longrightarrow U_S / \overline{\mathcal{O}_{K,+}^{\times}} \longrightarrow G_S \longrightarrow \text{Cl}_S^+ \longrightarrow 1.$$

Proof. An element of $K_{S,+}^\times$ lying in U_S has trivial divisor, hence is a totally positive unit. Applying div gives the surjection of G_S onto $I_S/P_{S,+}$, where $P_{S,+}$ is the group of principal fractional ideals with a totally positive generator supported on S ; by definition $I_S/P_{S,+} = \text{Cl}_S^+$, a finite group. The kernel is $U_S/(U_S \cap \overline{K_{S,+}^\times})$, and since U_S is open in $\mathbb{I}_S$ one has $U_S \cap \overline{K_{S,+}^\times} = U_S \cap K_{S,+}^\times = \mathcal{O}_{K,+}^\times$. Compactness follows since U_S is compact and Cl_S^+ finite. $\square$

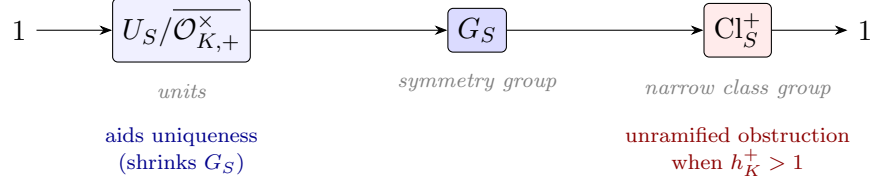


FIGURE 1. The two structures in (2.1) that are absent over $\mathbb{Q}$, where both ends are trivial and $G_S = \widehat{\mathbb{Z}}_S^\times$. The class group contributes obstructing characters (§4.1); the units act in the opposite direction (§4.3).

2.2. The algebra and its dynamics. Ideals of $\mathcal{O}_K$ need not be principal, and principal ideals have no canonical generator; the action of J_S is therefore defined through ideles, with the ambiguity absorbed by a balanced product.

Definition 2.6. Let $Y_S = \widehat{\mathcal{O}}_S \times_{U_S} G_S$ be the quotient of $\widehat{\mathcal{O}}_S \times G_S$ by $(x, g) \sim (wx, r(w)^{-1}g)$ for $w \in U_S$; it is a compact metrizable space. For $\mathfrak{a} \in J_S$ choose $s \in \mathbb{I}_S$ with $\text{div}(s) = \mathfrak{a}$ and $s_{\mathfrak{p}} \in \mathcal{O}_{\mathfrak{p}}$ for all $\mathfrak{p}$, and set

$$\mathfrak{a} \cdot [x, g] = [sx, r(s)^{-1}g],$$

independent of the choice of s because two choices differ by an element of U_S . This is a right action of the abelian monoid J_S by injective continuous maps with clopen images. The S -local Bost–Connes system is the semigroup crossed product

$$\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S = \overline{\text{span}}\{\mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^* : \mathfrak{a}, \mathfrak{b} \in J_S, f \in C(Y_S)\},$$

generated by $C(Y_S)$ and isometries $\mu_{\mathfrak{a}}$ with $\mu_{\mathfrak{a}} \mu_{\mathfrak{b}} = \mu_{\mathfrak{a}\mathfrak{b}}$, $\mu_{\mathfrak{a}} f \mu_{\mathfrak{a}}^* = f \circ (\mathfrak{a} \cdot)^{-1}$ extended by zero, $\mu_{\mathfrak{a}} \mu_{\mathfrak{a}}^* = \mathbf{1}_{\mathfrak{a}Y_S}$ and $\mu_{\mathfrak{a}} \mu_{\mathfrak{b}}^* = \mu_{\mathfrak{b}}^* \mu_{\mathfrak{a}}$ for $\mathfrak{a}, \mathfrak{b}$ coprime. The dynamics is

$$\sigma_t(f) = f, \quad \sigma_t(\mu_{\mathfrak{a}}) = \text{Na}^{it} \mu_{\mathfrak{a}}.$$

The group G_S acts by $\theta_h[x, g] = [x, gh^{-1}]$, commuting with σ .

For $K = \mathbb{Q}$ this is the system of [12] restricted to S ; for $S = P_K$ it is the system of [7, 11]. We record the inductive limit structure, which is what allows sparse sets to be approached by finite ones.

Proposition 2.7. *If $S = \bigcup_k S_k$ with $S_1 \subseteq S_2 \subseteq \dots$, then the maps $f \mapsto f \circ \text{pr}$, $\mu_{\mathfrak{a}} \mapsto \mu_{\mathfrak{a}}$ define unital σ -equivariant embeddings $\mathcal{A}_{K,S_k} \hookrightarrow \mathcal{A}_{K,S_{k+1}}$ and $\varinjlim_k \mathcal{A}_{K,S_k} \cong \mathcal{A}_{K,S}$ equivariantly for $G_S = \varinjlim_k G_{S_k}$. In particular $\mathcal{A}_{K,S} \subseteq \mathcal{A}_{K,P_K}$ for every S .*

Proof. The relations are preserved because $\mathbf{1}_{\mathfrak{a}Y_{S'}}$ pulls back to $\mathbf{1}_{\mathfrak{a}Y_S}$ for $\mathfrak{a} \in J_S$; injectivity follows from the gauge-invariant uniqueness theorem, and surjectivity onto a dense subalgebra from $\widehat{\mathcal{O}}_S = \varinjlim \widehat{\mathcal{O}}_{S_k}$ together with $J_S = \bigcup_k J_{S_k}$ and Stone–Weierstrass. Strong continuity of the limit dynamics holds on the dense union and hence everywhere. $\square$

2.3. KMS states are measures. Let φ be a (σ, β) -KMS state. Over $\mathbb{Q}$ one shows $\varphi(\mu_m f \mu_n^*) = 0$ for $m \neq n$ by comparing m^β with n^β ; over a number field this argument is unavailable, because distinct ideals frequently have equal norms. We use instead the groupoid description.

Proposition 2.8. *Let $\beta > 0$ and set*

$$M_\beta(K, S) = \{\nu \in \text{Prob}(Y_S) : \nu(\mathbf{a} \cdot E) = N\mathbf{a}^{-\beta} \nu(E) \quad \forall \mathbf{a} \in J_S, E \text{ Borel}\}.$$

Then $\varphi \mapsto \varphi|_{C(Y_S)}$ is an affine bijection from the KMS_β simplex of $\mathcal{A}_{K,S}$ onto $M_\beta(K, S)$. There are no KMS_β states for $\beta < 0$, and for $\beta = 0$ the tracial states correspond to J_S -invariant measures.

Proof. J_S is abelian and cancellative, hence an Ore semigroup with enveloping group I_S . By Lacas's dilation theorem [10] the semigroup crossed product $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$ is a full corner in the group crossed product $C_0(\tilde{Y}_S) \rtimes I_S$, where $\tilde{Y}_S = \varinjlim (Y_S, \mathbf{a} \cdot)$ is the direct limit along the directed system J_S and $C(Y_S) = \mathbf{1}_{Y_S} C_0(\tilde{Y}_S) \mathbf{1}_{Y_S}$; the dilation is equivariant for the dynamics, which on the dilation is implemented by the $\mathbb{R}$ -valued cocycle $c(\mathbf{a}\mathbf{b}^{-1}) = \log N\mathbf{a} - \log N\mathbf{b}$ on I_S . Corner inclusions preserve KMS simplices [10, §3], so it suffices to treat $C_0(\tilde{Y}_S) \rtimes I_S = C^*(\mathcal{G}_S)$, the C^* -algebra of the transformation groupoid $\mathcal{G}_S = I_S \times \tilde{Y}_S$, which is étale, second countable and locally compact.

By Neshveyev's theorem [14, Thm. 1.3], KMS_β states of $C^*(\mathcal{G}_S)$ for the dynamics induced by c are in bijection with pairs consisting of a probability measure ν on $\tilde{Y}_S^{(0)}$ that is quasi-invariant with Radon–Nikodym cocycle $e^{-\beta c}$, together with a ν -measurable field of states on the C^* -algebras of the isotropy groups. Quasi-invariance with that cocycle is exactly $\nu(\mathbf{a}E) = N\mathbf{a}^{-\beta} \nu(E)$ for $\mathbf{a} \in J_S$.

It remains to see that the isotropy contributes nothing. Taking $E = \mathbf{a} \cdot Y_S$ gives $\nu(\mathbf{a}Y_S) = N\mathbf{a}^{-\beta}$; applying this to $\mathbf{a} = \mathbf{p}^k$ and letting $k \rightarrow \infty$ gives $\nu(\{x_{\mathbf{p}} = 0\}) = 0$ for each $\mathbf{p} \in S$, hence, S being countable, ν is concentrated on the set Y_S° of classes $[x, g]$ with all $x_{\mathbf{p}} \neq 0$. On Y_S° the action of I_S is free: by Lemma 2.10(2) below, in the coordinates $(v, g) \in \mathbb{Z}^S \times G_S$ an element $\mathbf{a}\mathbf{b}^{-1}$ acts by $v \mapsto v + \mathbf{a} - \mathbf{b}$, and $\mathbf{a} = \mathbf{b}$ forces $\mathbf{a} = \mathbf{b}$. So $\mathcal{G}_S$ is principal on a conull set, all isotropy groups are trivial there, and the field of states is empty data. The correspondence is affine by construction, and the state is recovered from ν by $\varphi = \nu \circ E_S$ with E_S the canonical conditional expectation onto $C(Y_S)$ given by averaging over the gauge action of $\widehat{I}_S$.

For $\beta < 0$: taking $\mathbf{a} \neq (1)$, $\nu(\mathbf{a}Y_S) = N\mathbf{a}^{-\beta} > 1$, impossible. $\square$

Remark 2.9. Concretely, $\varphi = \nu \circ E_S$ means $\varphi(\mu_{\mathbf{a}} f \mu_{\mathbf{b}}^*) = 0$ for $\mathbf{a} \neq \mathbf{b}$ and $\varphi(\mu_{\mathbf{a}} f \mu_{\mathbf{a}}^*) = \nu(f \circ (\mathbf{a} \cdot)^{-1})$. The vanishing for $\mathbf{a} \neq \mathbf{b}$ with $N\mathbf{a} = N\mathbf{b}$, which is invisible to the modular argument, is here a consequence of freeness of the I_S -action, i.e. of the fact that ideals, not their norms, index the translations.

2.4. Coordinates. Choose a uniformizer $\pi_{\mathbf{p}}$ at each $\mathbf{p} \in S$. Every $x \in \widehat{\mathcal{O}}_S$ with all coordinates nonzero factors uniquely as $x_{\mathbf{p}} = \pi_{\mathbf{p}}^{v_{\mathbf{p}}} u_{\mathbf{p}}$ with $v_{\mathbf{p}} \in \mathbb{Z}_{\geq 0}$, $u_{\mathbf{p}} \in \mathcal{O}_{\mathbf{p}}^\times$. Put $V = \mathbb{Z}_{\geq 0}^S$. For $\mathbf{a} = \prod \mathbf{p}^{a_{\mathbf{p}}} \in J_S$ let $s_{\mathbf{a}} = (\pi_{\mathbf{p}}^{a_{\mathbf{p}}})_{\mathbf{p}} \in \mathbb{I}_S$ and

$$(2.2) \quad \sigma_{\mathbf{a}} := r(s_{\mathbf{a}})^{-1} \in G_S.$$

- Lemma 2.10.** (1) $\mathbf{a} \mapsto \sigma_{\mathbf{a}}$ is a monoid homomorphism $J_S \rightarrow G_S$ extending to $\sigma : I_S \rightarrow G_S$; its composition with $G_S \rightarrow \text{Cl}_S^+$ is the inverse of the natural surjection $I_S \rightarrow \text{Cl}_S^+$, i.e. $\sigma_{\mathbf{p}} \mapsto [\mathbf{p}]^{-1}$, the inversion being that of (2.2). If $\mathbf{a} = (\alpha)$ with $\alpha \in K_{S,+}^\times$ then $\sigma_{\mathbf{a}} = r(\theta_\alpha)^{-1}$ with $\theta_\alpha := \alpha^{-1} s_{\mathbf{a}} \in U_S$; for $K = \mathbb{Q}$, $\sigma_\ell = \theta_\ell$, where $(\theta_\ell)_{\ell'} = \ell$ for $\ell' \neq \ell$ and 1 for $\ell' = \ell$.
- (2) $[x, g] \mapsto (v(x), r(u(x))g)$ is a Borel isomorphism from Y_S° onto $V \times G_S$, and in these coordinates $\mathbf{a} \cdot (v, g) = (v + \mathbf{a}, \sigma_{\mathbf{a}} g)$.
- (3) For every $\nu \in M_\beta(K, S)$ the law π_β of v is the same: the $v_{\mathbf{p}}$ are independent with $\pi_\beta(v_{\mathbf{p}} \geq k) = N\mathbf{p}^{-k\beta}$, so $\pi_\beta(v_{\mathbf{p}} = k) = (1 - N\mathbf{p}^{-\beta})N\mathbf{p}^{-k\beta}$.

- (4) If $(\rho_v)_{v \in V}$ is a disintegration of ν over π_β , then $\nu \in M_\beta(K, S)$ if and only if for every $\mathbf{a} \in J_S$

$$(2.3) \quad \rho_{v+a} = (\sigma_{\mathbf{a}})_* \rho_v \quad \text{for } \pi_\beta\text{-a.e. } v.$$

Proof. (1) $s_{\mathbf{a}\mathbf{b}} = s_{\mathbf{a}}s_{\mathbf{b}}$, and r is a homomorphism. Composing with div modulo $P_{S,+}$ sends $s_{\mathbf{a}}$ to $[\mathbf{a}]$, hence $\sigma_{\mathbf{a}}$ to $[\mathbf{a}]^{-1}$. If $\mathbf{a} = (\alpha)$ with α totally positive then $\alpha^{-1}s_{\mathbf{a}}$ has trivial divisor, so lies in U_S , and $r(\alpha) = 1$ gives $r(s_{\mathbf{a}}) = r(\alpha^{-1}s_{\mathbf{a}})$. For $K = \mathbb{Q}$ take $\pi_\ell = \ell$ and $\alpha = \ell$: then θ_α has component ℓ^{-1} at every $\ell' \neq \ell$ and 1 at ℓ , so $\sigma_\ell = r(\theta_\alpha)^{-1} = \theta_\ell$.

(2) Well-definedness: replacing (x, g) by $(wx, r(w)^{-1}g)$ leaves v unchanged and replaces u by wu , so $r(u)g$ is unchanged, G_S being abelian. The inverse map is $(v, g) \mapsto [(\pi_{\mathbf{p}}^{v_{\mathbf{p}}})_{\mathbf{p}}, g]$. With $s = s_{\mathbf{a}}$ one has $v \mapsto v + a$, $u \mapsto u$ and $g \mapsto r(s_{\mathbf{a}})^{-1}g = \sigma_{\mathbf{a}}g$.

(3) For finite $F \subset S$ and $k_{\mathbf{p}} \geq 0$, with $\mathbf{a} = \prod_{\mathbf{p} \in F} \mathbf{p}^{k_{\mathbf{p}}}$, one has $\{v_{\mathbf{p}} \geq k_{\mathbf{p}} \forall \mathbf{p} \in F\} = \mathbf{a} \cdot Y_S$ modulo the null set $Y_S \setminus Y_S^\circ$, and $\nu(\mathbf{a}Y_S) = N\mathbf{a}^{-\beta} = \prod_{\mathbf{p} \in F} N\mathbf{p}^{-k_{\mathbf{p}}\beta}$. This gives independence and the marginals, hence π_β .

(4) Translation by a multiplies π_β by the constant $N\mathbf{a}^{-\beta}$ by (3). For product sets $A \times B$,

$$\nu(\mathbf{a} \cdot (A \times B)) = \int_{A+a} \rho_v(\sigma_{\mathbf{a}}B) d\pi_\beta(v), \quad N\mathbf{a}^{-\beta}\nu(A \times B) = \int_{A+a} \rho_{v-a}(B) d\pi_\beta(v),$$

the second by the change of variables $v \mapsto v - a$. As A, B are arbitrary, equality for all of them is (2.3); the converse is the same computation with Fubini. $\square$

3. THE CLASSIFICATION AND THE CHEBOTAREV CRITERION

Throughout this section K, S and $\beta > 0$ are fixed, and we work in the coordinates of Lemma 2.10. Write $N_{\mathbf{p}} := N\mathbf{p}$, $t_{\mathbf{p}} := N\mathbf{p}^{-\beta} \in (0, 1)$, and for $\chi \in \widehat{G_S}$

$$f_\chi(v) := \widehat{\rho}_v(\chi) = \int_{G_S} \overline{\chi(g)} d\rho_v(g), \quad |f_\chi| \leq 1.$$

By (2.3), for every $\mathbf{a} \in J_S$ with valuation vector a ,

$$(3.1) \quad f_\chi(v + a) = \overline{\chi(\sigma_{\mathbf{a}})} f_\chi(v) \quad \pi_\beta\text{-a.e.}$$

Set $c_{\mathbf{p}} := \overline{\chi(\sigma_{\mathbf{p}})} \in \mathbb{T}$, so that $c^{\mathbf{a}} = \prod_{\mathbf{p}} c_{\mathbf{p}}^{a_{\mathbf{p}}} = \overline{\chi(\sigma_{\mathbf{a}})}$.

Lemma 3.1 (Zero-one law). $|f_\chi|$ is π_β -a.e. equal to a constant $c_\chi \in [0, 1]$.

Proof. Taking $\mathbf{a} = \mathbf{p}$ in (3.1) gives $|f_\chi|(v + e_{\mathbf{p}}) = |f_\chi|(v)$ a.e. The orbits of $v \mapsto v + e_{\mathbf{p}}$ are the fibres of the projection $V \rightarrow \mathbb{Z}_{\geq 0}^{S \setminus \{\mathbf{p}\}}$, so $|f_\chi|$ agrees a.e. with a function not depending on $v_{\mathbf{p}}$; that is, $|f_\chi|$ is measurable with respect to $\sigma(v_{\mathbf{p}'} : \mathbf{p}' \neq \mathbf{p})$ modulo null sets. Applying this coordinate by coordinate over a finite set F , $|f_\chi|$ is measurable with respect to $\sigma(v_{\mathbf{p}'} : \mathbf{p}' \notin F)$ for every finite F , hence with respect to the tail σ -algebra of the independent family $(v_{\mathbf{p}})_{\mathbf{p} \in S}$. By Kolmogorov's zero-one law that σ -algebra is π_β -trivial. $\square$

Suppose $\chi \neq 1$ and $c_\chi > 0$. Then $g := f_\chi/|f_\chi|$ is a measurable map $V \rightarrow \mathbb{T}$, defined a.e., with

$$(3.2) \quad g(v + e_{\mathbf{p}}) = c_{\mathbf{p}} g(v) \quad \pi_\beta\text{-a.e., for every } \mathbf{p} \in S.$$

Put

$$m_{\mathbf{p}} := \mathbb{E}_{\pi_\beta}[c_{\mathbf{p}}^{v_{\mathbf{p}}}] = \sum_{k \geq 0} (1 - t_{\mathbf{p}}) t_{\mathbf{p}}^k c_{\mathbf{p}}^k = \frac{1 - t_{\mathbf{p}}}{1 - c_{\mathbf{p}} t_{\mathbf{p}}} \in \mathbb{C}.$$

Lemma 3.2. $1 - |m_{\mathbf{p}}|^2 = \frac{2t_{\mathbf{p}}(1 - \operatorname{Re} c_{\mathbf{p}})}{|1 - c_{\mathbf{p}}t_{\mathbf{p}}|^2}$, and therefore

$$\sum_{\mathbf{p} \in S} (1 - |m_{\mathbf{p}}|) < \infty \iff \sum_{\mathbf{p} \in S} N_{\mathbf{p}}^{-\beta} |1 - \chi(\sigma_{\mathbf{p}})|^2 < \infty.$$

Proof. $|m_{\mathbf{p}}|^2 = (1 - t_{\mathbf{p}})^2 / |1 - c_{\mathbf{p}}t_{\mathbf{p}}|^2$ and $|1 - c_{\mathbf{p}}t_{\mathbf{p}}|^2 = 1 - 2t_{\mathbf{p}} \operatorname{Re} c_{\mathbf{p}} + t_{\mathbf{p}}^2$ give the identity. Since $|1 - c_{\mathbf{p}}|^2 = 2(1 - \operatorname{Re} c_{\mathbf{p}})$ and, for all but at most one $\mathbf{p}$ (the one of least norm), $t_{\mathbf{p}}$ is bounded away from 1 by a constant depending only on β and S , we get $1 - |m_{\mathbf{p}}|^2 \asymp t_{\mathbf{p}}|1 - c_{\mathbf{p}}|^2$ up to finitely many terms; and $1 - |m| \leq 1 - |m|^2 \leq 2(1 - |m|)$. $\square$

Proposition 3.3 (Necessity). *If $c_{\chi} > 0$ then $\sum_{\mathbf{p} \in S} (1 - |m_{\mathbf{p}}|) < \infty$, i.e. $\chi \in \Xi_{\beta}(K, S)$.*

Proof. Let g be as in (3.2), $|g| = 1$ a.e. Fix an increasing exhaustion $F_1 \subset F_2 \subset \dots$ of S by finite sets; write $\mathcal{F}_F = \sigma(v_{\mathbf{p}} : \mathbf{p} \in F)$ and $P_F = \prod_{\mathbf{p} \in F} c_{\mathbf{p}}^{v_{\mathbf{p}}}$.

Iterating (3.2) in the coordinate $\mathbf{p}$ shows that $g c_{\mathbf{p}}^{-v_{\mathbf{p}}}$ agrees a.e. with a function not depending on $v_{\mathbf{p}}$; doing this for all $\mathbf{p} \in F$ shows that $G_F := g P_F^{-1}$ is measurable with respect to $\sigma(v_{\mathbf{p}} : \mathbf{p} \notin F)$, hence independent of $\mathcal{F}_F$. Therefore

$$\mathbb{E}[g \mid \mathcal{F}_F] = P_F \mathbb{E}[G_F], \quad \text{so} \quad |\mathbb{E}[g \mid \mathcal{F}_F]| = |\mathbb{E}[G_F]|.$$

Since $\mathcal{F}_{F_k} \uparrow \sigma(v)$, martingale convergence gives $\mathbb{E}[g \mid \mathcal{F}_{F_k}] \rightarrow g$ in $L^2(\pi_{\beta})$, so $|\mathbb{E}[G_{F_k}]| \rightarrow |g| = 1$. For large k put $\gamma_k = \mathbb{E}[G_{F_k}] / |\mathbb{E}[G_{F_k}]|$. Then, as $|P_{F_k}| = 1$,

$$\|g - \gamma_k P_{F_k}\|_2^2 = \|G_{F_k} - \gamma_k\|_2^2 = 2 - 2|\mathbb{E}[G_{F_k}]| \rightarrow 0.$$

Hence $(\gamma_k P_{F_k})_k$ is Cauchy in L^2 . For $j < k$, dividing by the unimodular P_{F_j} and using independence,

$$\|\gamma_k P_{F_k} - \gamma_j P_{F_j}\|_2 = \left\| \gamma_k \prod_{\mathbf{p} \in F_k \setminus F_j} c_{\mathbf{p}}^{v_{\mathbf{p}}} - \gamma_j \right\|_2 \geq 1 - \left| \mathbb{E} \prod_{\mathbf{p} \in F_k \setminus F_j} c_{\mathbf{p}}^{v_{\mathbf{p}}} \right| = 1 - \prod_{\mathbf{p} \in F_k \setminus F_j} |m_{\mathbf{p}}|,$$

where the inequality is $\|X - \gamma\|_2 \geq |\mathbb{E}X - \gamma| \geq 1 - |\mathbb{E}X|$ for unimodular constants γ . Thus $\prod_{\mathbf{p} \in F_k \setminus F_j} |m_{\mathbf{p}}| \rightarrow 1$ as $j, k \rightarrow \infty$, the Cauchy criterion for convergence of $\prod_{\mathbf{p} \in S} |m_{\mathbf{p}}|$ to a nonzero limit, i.e. $\sum_{\mathbf{p}} (1 - |m_{\mathbf{p}}|) < \infty$. Lemma 3.2 concludes. $\square$

Proposition 3.4 (Sufficiency). *If $\chi \in \Xi_{\beta}(K, S)$ then there is a measurable $g_{\chi} : V \rightarrow \mathbb{T}$ with $|g_{\chi}| = 1$ satisfying (3.2), unique up to a multiplicative constant. If moreover $\chi \neq 1$, there is $\nu \in M_{\beta}(K, S)$ that is not the symmetric measure; in particular the $\operatorname{KMS}_{\beta}$ state is not unique.*

Proof. Set $\tilde{X}_{\mathbf{p}} = \overline{(m_{\mathbf{p}}/|m_{\mathbf{p}}|)} c_{\mathbf{p}}^{v_{\mathbf{p}}}$, well defined because $m_{\mathbf{p}} \neq 0$; these are independent, unimodular, with $\mathbb{E}\tilde{X}_{\mathbf{p}} = |m_{\mathbf{p}}| > 0$. Put $\tilde{P}_k = \prod_{\mathbf{p} \in F_k} \tilde{X}_{\mathbf{p}}$ and $L_k = \prod_{\mathbf{p} \in F_k} |m_{\mathbf{p}}|$; by hypothesis and Lemma 3.2, $L_k \downarrow L > 0$. Then $N_k := \tilde{P}_k / L_k$ is an $(\mathcal{F}_{F_k})$ -martingale with $\|N_k\|_2 = 1/L_k \leq 1/L$, hence converges a.s. and in L^2 ; so $\tilde{P}_k \rightarrow g_{\chi}$ a.s. with $|g_{\chi}| = 1$, and passing to the limit in $\tilde{P}_k(v + e_{\mathbf{p}}) = c_{\mathbf{p}} \tilde{P}_k(v)$ (valid once $\mathbf{p} \in F_k$) gives (3.2). If g, g' both satisfy (3.2) then g/g' is invariant under all $v \mapsto v + e_{\mathbf{p}}$, hence constant by Lemma 3.1.

Now let $\chi \neq 1$ and define ρ_v to be the measure on G_S with density $1 + \operatorname{Re}(g_{\chi}(v)\chi(g)) \geq 0$ with respect to normalized Haar measure. It is a probability measure whose Fourier coefficients $\hat{\rho}_v(\chi') = \int \chi' d\rho_v$ are $\hat{\rho}_v(1) = 1$, $\hat{\rho}_v(\chi) = \frac{1}{2}g_{\chi}(v)$, $\hat{\rho}_v(\bar{\chi}) = \frac{1}{2}\overline{g_{\chi}(v)}$ and 0 elsewhere when $\chi^2 \neq 1$; when $\chi^2 = 1$ the two coefficients merge into $\hat{\rho}_v(\chi) = \operatorname{Re} g_{\chi}(v) = g_{\chi}(v)$, which is real because then all $c_{\mathbf{p}} = \pm 1$ and all $m_{\mathbf{p}} > 0$, so that $g_{\chi} = \lim_k \prod_{\mathbf{p} \in F_k} c_{\mathbf{p}}^{v_{\mathbf{p}}} \in \{\pm 1\}$. Since $\widehat{(\sigma_a)_* \rho_v}(\chi') = \overline{\chi'(\sigma_a)} \hat{\rho}_v(\chi')$, comparison with $\widehat{\rho_{v+a}}(\chi')$ coefficient by coefficient reduces to (3.2) for $\chi' = \chi$, to its conjugate for $\chi' = \bar{\chi}$, and to $1 = 1$ or $0 = 0$ otherwise. So (2.3) holds and Lemma 2.10(4) yields

$\nu \in M_\beta(K, S)$, with $\widehat{\rho}_\nu(\chi) \neq 0$, whereas the symmetric measure has ρ_ν Haar and all nontrivial coefficients zero. $\square$

Lemma 3.5. $\Xi_\beta(K, S)$ is a subgroup of $\widehat{G}_S$; it equals $\widehat{G}_S$ whenever $\zeta_{K,S}(\beta) < \infty$, and $\beta \mapsto \Xi_\beta$ is non-decreasing.

Proof. $|1 - \chi_1\chi_2(\sigma)| \leq |1 - \chi_1(\sigma)| + |1 - \chi_2(\sigma)|$ and $(a+b)^2 \leq 2a^2 + 2b^2$ give closure under products; $|1 - \bar{\chi}(\sigma)| = |1 - \chi(\sigma)|$ under inverses. If $\sum_{\mathfrak{p}} N\mathfrak{p}^{-\beta} < \infty$ every χ qualifies since $|1 - \chi(\sigma_{\mathfrak{p}})| \leq 2$. Monotonicity holds because $N\mathfrak{p}^{-\beta'} \leq N\mathfrak{p}^{-\beta}$ for $\beta \leq \beta'$. $\square$

Theorem 3.6 (Classification). *Let K be a number field, S a nonempty set of finite primes and $\beta > 0$. Then the KMS_β simplex of $\mathcal{A}_{K,S}$ is affinely isomorphic to*

$$\text{Prob}(G_S/\Xi_\beta(K, S)^\perp), \quad \Xi_\beta^\perp = \{g \in G_S : \chi(g) = 1 \ \forall \chi \in \Xi_\beta\}.$$

In particular:

- (1) the KMS_β state is unique if and only if $\Xi_\beta(K, S) = \{1\}$;
- (2) if $\zeta_{K,S}(\beta) < \infty$ then $\Xi_\beta = \widehat{G}_S$, $\Xi_\beta^\perp = \{1\}$, the simplex is $\text{Prob}(G_S)$, its extreme points are the Gibbs states of the Fock representations, of type I_∞ , and G_S permutes them freely and transitively;
- (3) the symmetry G_S acts on the simplex through the quotient $G_S/\Xi_\beta^\perp$, so $\Xi_\beta^\perp$ is the residual symmetry group of the phase.

Proof. By Proposition 2.8 and Lemma 2.10, a KMS state is determined by the family (ρ_v) , hence by $(f_\chi)_{\chi \in \widehat{G}_S}$. By Lemma 3.1 and Proposition 3.3, $f_\chi \equiv 0$ for $\chi \notin \Xi_\beta$. For $\chi \in \Xi_\beta$ let g_χ be a solution of (3.2) with $|g_\chi| = 1$, unique up to a unimodular constant by Proposition 3.4. The constants can be chosen so that $\chi \mapsto g_\chi(v)$ is multiplicative. Indeed, for $\chi_1, \chi_2 \in \Xi_\beta$ the unimodular function $g_{\chi_1}g_{\chi_2}/g_{\chi_1\chi_2}$ satisfies (3.2) with every $c_{\mathfrak{p}}$ replaced by 1, hence is invariant under all the shifts $v \mapsto v + e_{\mathfrak{p}}$ and is a.e. equal to a constant $\omega(\chi_1, \chi_2) \in \mathbb{T}$ by the argument of Lemma 3.1. The function ω is a symmetric 2-cocycle on the countable abelian group Ξ_β with values in the divisible group $\mathbb{T}$; symmetric cocycles classify abelian extensions, and $\text{Ext}(\Xi_\beta, \mathbb{T}) = 0$, so ω is a coboundary, $\omega(\chi_1, \chi_2) = \lambda(\chi_1)\lambda(\chi_2)/\lambda(\chi_1\chi_2)$ for some $\lambda : \Xi_\beta \rightarrow \mathbb{T}$. Replacing g_χ by $g_\chi/\lambda(\chi)$ gives $g_{\chi_1}g_{\chi_2} = g_{\chi_1\chi_2}$, whence also $g_1 = 1$ and $g_{\bar{\chi}} = \overline{g_\chi}$. (The explicit normalization $\lim_k \prod_{\mathfrak{p} \in F_k} (m_{\mathfrak{p}}/|m_{\mathfrak{p}}|)c_{\mathfrak{p}}^{v_{\mathfrak{p}}}$ of Proposition 3.4 is multiplicative only up to such a cocycle, the arguments of the $m_{\mathfrak{p}}$ not being additive in χ .) Then $f_\chi = \kappa(\chi)g_\chi$ for a scalar $\kappa(\chi)$ with $\kappa(1) = 1$, and κ is positive definite on the discrete group Ξ_β : for $\chi_i \in \Xi_\beta$, $z_i \in \mathbb{C}$,

$$\sum_{i,j} z_i \bar{z}_j \kappa(\chi_i \bar{\chi}_j) = \mathbb{E}_{\pi_\beta} \left[\int_{G_S} \left| \sum_i z_i \overline{\chi_i(g)} g_{\chi_i}(v) \right|^2 d\rho_v(g) \right] \geq 0.$$

By Bochner's theorem κ is the Fourier transform of a unique probability measure on the compact dual $\widehat{\Xi_\beta} \cong G_S/\Xi_\beta^\perp$. Conversely every such measure arises: given κ positive definite with $\kappa(1) = 1$, define ρ_v by $\widehat{\rho}_v(\chi) = \kappa(\chi)g_\chi(v)$ for $\chi \in \Xi_\beta$ and 0 otherwise; positivity of ρ_v is Bochner again (the function $\chi \mapsto \kappa(\chi)g_\chi(v)$ is positive definite for each fixed v , being κ twisted by the character $\chi \mapsto g_\chi(v)$ of Ξ_β), and (2.3) holds by (3.2). The correspondence $\nu \leftrightarrow \kappa$ is affine and bijective.

(1) and (3) are immediate. For (2), $\Xi_\beta = \widehat{G}_S$ by Lemma 3.5, so the simplex is $\text{Prob}(G_S)$; the extreme points correspond to point masses, and unwinding the correspondence they are the states $\zeta_{K,S}(\beta)^{-1} \text{Tr}(\pi_g(\cdot)e^{-\beta H})$ in the Fock representations π_g on $\ell^2(J_S)$ determined by $g \in G_S$, which are irreducible; the action of G_S is translation on the parameter. $\square$

Remark 3.7. Theorem 3.6 is a single statement covering both classical regimes. The high-temperature regime is $\Xi_\beta = \{1\}$ and the low-temperature regime $\Xi_\beta = \widehat{G}_S$; for the classical

systems these are the only two values that occur, but §4.2 shows that intermediate values do occur, giving phases with partial symmetry breaking that have no analogue in [1, 11].

3.1. The Chebotarev criterion.

Lemma 3.8. *Let $\chi \in \widehat{G_S}$ be nontrivial. Then χ has finite order and is trivial on an open subgroup, so there are a finite $F \subset S$ and exponents $k_{\mathfrak{p}} \geq 1$ with χ trivial on $\prod_{\mathfrak{p} \in F} (1 + \mathfrak{p}^{k_{\mathfrak{p}}} \mathcal{O}_{\mathfrak{p}}) \times \prod_{\mathfrak{p} \in S \setminus F} \mathcal{O}_{\mathfrak{p}}^{\times}$; put $\mathfrak{m} = \prod_{\mathfrak{p} \in F} \mathfrak{p}^{k_{\mathfrak{p}}}$. Let L/K be the finite cyclic extension attached by class field theory to an extension of χ to $\text{Gal}(K^{ab}/K)$ trivial on the inertia groups at all $\mathfrak{p} \notin S$; such extensions exist, and the statements below do not depend on the choice (see the proof). Then L lies in the ray class field of conductor dividing $\mathfrak{m}\infty$, hence is unramified outside $S \cup \infty$, and for every $\mathfrak{p} \in S$ with $\mathfrak{p} \nmid \mathfrak{m}$,*

$$\chi(\sigma_{\mathfrak{p}}) = \chi(\text{Frob}_{\mathfrak{p}})^{-1}.$$

In particular $\chi(\sigma_{\mathfrak{p}}) = 1$ if and only if $\mathfrak{p}$ splits completely in L .

Proof. First, how G_S sits inside the Galois group. The inclusion $\mathbb{I}_S \subset \mathbb{A}_{K,f}^{\times}$ (ideles with component 1 outside S) composed with the Artin map of Lemma 2.3 gives a continuous homomorphism $G_S \rightarrow \text{Gal}(K^{ab}/K)$, and its composite with the projection onto $\text{Gal}(K^S/K)$, where K^S is the maximal abelian extension of K unramified outside $S \cup \infty$, is injective: if $\alpha_n w_n \rightarrow s \in \mathbb{I}_S$ with $\alpha_n \in K_+^{\times}$ and $w_n \in \prod_{\mathfrak{p} \notin S} \mathcal{O}_{\mathfrak{p}}^{\times}$, then, the topology of $\mathbb{A}_{K,f}^{\times}$ being that of the restricted product, α_n is eventually a unit at every $\mathfrak{p} \notin S$, so $\alpha_n \in K_{S,+}^{\times}$ and $s \in \overline{K_{S,+}^{\times}}$. Thus G_S is a closed subgroup of $\text{Gal}(K^S/K)$, and χ extends to a character of $\text{Gal}(K^{ab}/K)$ trivial on the inertia groups at all $\mathfrak{p} \notin S$. Different extensions define different fields L , but for $\mathfrak{p} \in S$ the element $\text{Frob}_{\mathfrak{p}}$ is the image of $s_{\mathfrak{p}} \in \mathbb{I}_S$, so $\chi(\text{Frob}_{\mathfrak{p}})$ does not depend on the extension.

$\widehat{G_S}$ is torsion because G_S is a quotient of the totally disconnected group $\mathbb{I}_S$ by a closed subgroup, hence profinite. With the arithmetic normalization of the Artin map, the idele $s_{\mathfrak{p}}$ (a uniformizer at $\mathfrak{p}$, 1 elsewhere) maps to $\text{Frob}_{\mathfrak{p}}$ for $\mathfrak{p}$ unramified in L ; by (2.2), $\sigma_{\mathfrak{p}} = r(s_{\mathfrak{p}})^{-1}$, whence the displayed formula. For L/K abelian, $\text{Frob}_{\mathfrak{p}} = 1$ if and only if $\mathfrak{p}$ splits completely.

Two remarks on the archimedean places. First, G_S is a quotient by the *totally positive* group $K_{S,+}^{\times}$, so every character of G_S is trivial on totally positive elements and the sign conditions at the real places are already absorbed: the conductor attached to χ is $\mathfrak{m}\infty$ with the full archimedean part, and L lies in the *narrow* ray class field of that conductor. This is the same choice that forces the narrow class group in (2.1), and it is dictated by Lemma 2.3. Second, no archimedean condition enters (3.3), which involves only finite primes; the archimedean part determines which fields L occur, not the series. $\square$

Theorem 3.9 (Chebotarev criterion). *Let $\beta > 0$. The KMS_{β} state of $\mathcal{A}_{K,S}$ is unique if and only if*

$$(3.3) \quad \sum_{\substack{\mathfrak{p} \in S \\ \mathfrak{p} \text{ not split completely in } L}} N\mathfrak{p}^{-\beta} = \infty$$

for every finite abelian extension L/K inside K^{ab} , unramified outside $S \cup \infty$, whose associated character of G_S is nontrivial.

Proof. By Theorem 3.6(1) uniqueness means $\Xi_{\beta} = \{1\}$. By Lemma 3.8, for nontrivial χ the quantity $|1 - \chi(\sigma_{\mathfrak{p}})|^2$ vanishes exactly on the primes of S splitting completely in L and is at least $\min_{\omega \neq 1} |1 - \omega|^2 > 0$ elsewhere, ω ranging over the finitely many values of χ ; the finitely many $\mathfrak{p} \mid \mathfrak{m}$ do not affect convergence. Hence $\chi \notin \Xi_{\beta}$ is exactly (3.3). $\square$

Corollary 3.10 (The classical systems). *Let $S = P_K$ and $0 < \beta \leq 1$. Then $\Xi_\beta(K, P_K) = \{1\}$, so the KMS_β state of the Bost–Connes system of K is unique. For $\beta > 1$, $\zeta_K(\beta) < \infty$ and Theorem 3.6(2) gives the extremal Gibbs states permuted freely and transitively by $\text{Gal}(K^{ab}/K)$.*

Proof. Let L/K be nontrivial abelian of degree $m \geq 2$. By the Chebotarev density theorem the primes of K not splitting completely in L have natural density $1 - 1/m \geq \frac{1}{2}$, so $\sum \text{Np}^{-1} = \infty$ over that set, hence $\sum \text{Np}^{-\beta} = \infty$ for $\beta \leq 1$. Apply Theorem 3.9. The second statement uses $\zeta_K(\beta) < \infty$ for $\beta > 1$ and Corollary 2.4. $\square$

Remark 3.11 (L -functions). For a nontrivial ray class character $\chi \bmod \mathfrak{m}$ and $\beta > 1$,

$$\log \zeta_{K,S}(\beta) - \text{Re} \log L_S(\beta, \chi) = \sum_{\mathfrak{p} \in S} \text{Np}^{-\beta} (1 - \text{Re} \chi(\mathfrak{p})) + O(1),$$

the error absorbing higher prime powers. So $\chi \in \Xi_\beta$ says exactly that $\log |L_S(\beta, \chi)|$ tracks $\log \zeta_{K,S}(\beta)$ to within a bounded error. For $S = P_K$ and $\beta \downarrow 1$ the left side tends to $+\infty$, because ζ_K has a simple pole at $\beta = 1$ while $L(\beta, \chi) \rightarrow L(1, \chi) \neq 0$. Thus the uniqueness half of the Bost–Connes theorem for K is, in this formulation, precisely the non-vanishing $L(1, \chi) \neq 0$ of Hecke [4] and Landau [5] (see also [15, Ch. VII]); over $\mathbb{Q}$ it is Dirichlet’s theorem. Chebotarev enters as the density-theoretic form of the same input.

Remark 3.12 (Primes of degree ≥ 2 are negligible). At most $d = [K : \mathbb{Q}]$ primes lie above each rational prime, so $\sum_{\deg \mathfrak{p} \geq 2} \text{Np}^{-\beta} \leq d \sum_p p^{-2\beta} < \infty$ for $\beta > \frac{1}{2}$. Hence for $\beta > \frac{1}{2}$ the group $\Xi_\beta(K, S)$ is unchanged if S is replaced by its set of degree-one primes, and all obstructions in the range of interest are carried by primes with $\text{Np} = p$.

3.2. Failure of the ergodicity hypothesis. It is natural to hope that whenever $\zeta_{K,S}(\beta) = \infty$ —so that the units carry measure zero and no Gibbs state exists—the action of $K_{S,+}^\times$ on $\hat{\mathcal{O}}_S$ is ergodic for every $\nu \in M_\beta(K, S)$, whence uniqueness. This is how the classical uniqueness theorem is proved in [13]. Theorem 3.6 shows the hope is unfounded in general.

Example 3.13 (Non-uniqueness with divergent partition function). Let $K = \mathbb{Q}$ and $S = \{3\} \cup \{\ell \equiv 1 \pmod{3}\}$. Then $\sum_{\ell \in S} \ell^{-1} = \infty$, so $\zeta_S(\beta) = \infty$ for $\beta \leq 1$. Let χ be the quadratic character of $\mathbb{Z}_3^\times$, regarded as an element of $\widehat{G}_S = \widehat{\mathbb{Z}_S^\times}$ supported at the component 3. For $\ell \in S$, $\ell \neq 3$, one has $(\sigma_\ell)_3 = \ell \equiv 1 \pmod{3}$, so $\chi(\sigma_\ell) = 1$; and $(\sigma_3)_3 = 1$, so $\chi(\sigma_3) = 1$. Hence the sum in (1.1) vanishes identically and $\chi \in \Xi_\beta$ for every $\beta > 0$. By Theorem 3.6 the KMS_β state is never unique, although there are no Gibbs states for $\beta \leq 1$.

Here $\sigma(I_S)$ is contained in the proper closed subgroup $\ker \chi$, so the failure is already visible topologically. Section 4.2 gives examples where it is not.

Remark 3.14. The correct general statement replacing the ergodicity hypothesis is Theorem 3.9. What the classical proof uses is not the divergence of $\zeta_{K,S}$ but the equidistribution of S in ray classes; the two coincide for $S = P_K$ and diverge for sparse S .

4. ALGEBRAIC OBSTRUCTIONS AND THE SEPARATION OF TOPOLOGY AND MEASURE

Theorem 3.9 reduces uniqueness to a statement about how the primes of S are distributed among Frobenius classes. Two independent mechanisms can make it fail. The first is arithmetic and specific to fields of narrow class number > 1 ; the second is measure-theoretic and occurs already over $\mathbb{Q}$.

4.1. The unramified class-group obstruction. Over $\mathbb{Q}$ every nontrivial character of $G_S = \widehat{\mathbb{Z}}_S^\times$ is built from the components at finitely many primes of S : the modulus must be supported on S , and the associated field is ramified there. Over a field with $h_K^+ > 1$ this is no longer so, because (2.1) provides a surjection $G_S \twoheadrightarrow \text{Cl}_S^+$ whose characters are trivial on all local units.

Proposition 4.1. *Let ψ be a character of Cl_S^+ and $\chi = \psi \circ (G_S \rightarrow \text{Cl}_S^+)$. Then $\chi(\sigma_{\mathfrak{p}}) = \psi([\mathfrak{p}])^{-1} = \overline{\psi([\mathfrak{p}])}$ for every $\mathfrak{p} \in S$, the field L attached to χ is contained in the narrow Hilbert class field H^+ of K and is therefore unramified at all finite primes, and*

$$\chi \in \Xi_\beta(K, S) \iff \sum_{\substack{\mathfrak{p} \in S \\ [\mathfrak{p}] \notin \ker \psi}} N\mathfrak{p}^{-\beta} < \infty.$$

Consequently, if for some proper subgroup $H < \text{Cl}_S^+$ the primes of S lying outside H form a β -summable set, then the KMS_β state of $\mathcal{A}_{K,S}$ is not unique, and no ramified data is involved.

Proof. Lemma 2.10(1) identifies the image of $\sigma_{\mathfrak{p}}$ in Cl_S^+ with $[\mathfrak{p}]^{-1}$, and inversion does not affect $|1 - \chi(\sigma_{\mathfrak{p}})|$; the rest is Theorem 3.6 together with the computation in the proof of Theorem 3.9. Characters of Cl_K^+ correspond to subfields of H^+ by class field theory. $\square$

Example 4.2 ($K = \mathbb{Q}(\sqrt{-5})$). Here $h_K = h_K^+ = 2$ (there are no real places, so the narrow and wide class groups agree), $\mathcal{O}_K^\times = \{\pm 1\}$ is finite, and $H = K(i)$. For a rational prime $p \neq 2, 5$ split in K , the primes above p are principal iff $p = x^2 + 5y^2$ iff $p \equiv 1, 9 \pmod{20}$, and non-principal iff $2p = x^2 + 5y^2$ iff $p \equiv 3, 7 \pmod{20}$.

Let $S_0 = \{\mathfrak{p} : \deg \mathfrak{p} = 1, [\mathfrak{p}] = 1\}$ and let $R = \{\mathfrak{p}_1, \mathfrak{p}_2, \dots\}$ be an infinite set of degree-one primes with $[\mathfrak{p}] \neq 1$, chosen so sparsely that $N\mathfrak{p}_k > 2^{2^k}$; put $S = S_0 \cup R$. Then:

- (1) $\sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-1} = \infty$, since by Chebotarev the rational primes $\equiv 1, 9 \pmod{20}$ have positive density; hence $\beta_c(K, S) = 1$ and $\zeta_{K,S}(\beta) = \infty$ for $0 < \beta \leq 1$;
- (2) $\sum_{\mathfrak{p} \in R} N\mathfrak{p}^{-\beta} \leq \sum_k 2^{-\beta 2^k} < \infty$ for every $\beta > 0$;
- (3) with ψ the nontrivial character of $\text{Cl}_K \cong \mathbb{Z}/2\mathbb{Z}$ and $\chi = \psi \circ (G_S \rightarrow \text{Cl}_S)$ —note $\text{Cl}_S = \text{Cl}_K$ because $R \neq \emptyset$ —Proposition 4.1 gives $\sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}})|^2 = 4 \sum_{\mathfrak{p} \in R} N\mathfrak{p}^{-\beta} < \infty$, so $\chi \in \Xi_\beta(K, S)$ for every $\beta > 0$;
- (4) $\sigma(I_S)$ is dense in G_S (Proposition 4.3).

Hence for every $\beta \in (0, 1]$ the KMS_β simplex of $\mathcal{A}_{K,S}$ is a segment with exactly two extreme points, neither of them a Gibbs state, and G_S acts on them through $G_S / \ker \chi \cong \mathbb{Z}/2\mathbb{Z}$. The symmetry is partially broken at every temperature, and the obstruction is everywhere unramified.

Proposition 4.3. *With S as in Example 4.2, $\sigma(I_S)$ is dense in G_S .*

Proof. $G_S = \varprojlim_{\mathfrak{m}} G_S / r(W_{\mathfrak{m}})$ where $W_{\mathfrak{m}} = \prod_{\mathfrak{p} \in F} (1 + \mathfrak{p}^{k_{\mathfrak{p}}} \mathcal{O}_{\mathfrak{p}}) \times \prod_{\mathfrak{p} \in S \setminus F} \mathcal{O}_{\mathfrak{p}}^\times$ runs over the open subgroups of the above form, so it suffices to prove surjectivity onto each finite quotient $C_{\mathfrak{m}} := G_S / r(W_{\mathfrak{m}})$. By Definition 2.1 and (2.1), $C_{\mathfrak{m}}$ is a quotient of the ray class group $\text{Cl}_{\mathfrak{m}}(K)$ in which the image of $\sigma_{\mathfrak{p}}$ is the class of $\mathfrak{p}$ for $\mathfrak{p} \nmid \mathfrak{m}$; so we must show that $\{[\mathfrak{p}]_{C_{\mathfrak{m}}} : \mathfrak{p} \in S, \mathfrak{p} \nmid \mathfrak{m}\}$ generates $C_{\mathfrak{m}}$.

Let $C_{\mathfrak{m}}^\circ = \ker(C_{\mathfrak{m}} \rightarrow \text{Cl}_K)$, of index 1 or 2. By the Chebotarev density theorem applied to the ray class field of conductor $\mathfrak{m}$, every class of $C_{\mathfrak{m}}^\circ$ contains infinitely many degree-one primes; each such prime is principal in Cl_K , hence lies in $S_0 \subseteq S$, and all but finitely many avoid F . So the generated subgroup contains $C_{\mathfrak{m}}^\circ$. If $C_{\mathfrak{m}}^\circ = C_{\mathfrak{m}}$ we are done. Otherwise choose $\mathfrak{p} \in R \setminus F$, which exists as R is infinite; its class lies outside $C_{\mathfrak{m}}^\circ$ because $[\mathfrak{p}] \neq 1$ in Cl_K , and the index is 2, so the generated subgroup is all of $C_{\mathfrak{m}}$. $\square$

4.2. Density is not enough. We now work over $K = \mathbb{Q}$, where $G_S = \widehat{\mathbb{Z}}_S^\times$ and, by Lemma 2.10(1), $\sigma_\ell = \theta_\ell$ with $(\theta_\ell)_{\ell'} = \ell$ for $\ell' \neq \ell$ and 1 for $\ell' = \ell$; we write $\Theta_S = \sigma(I_S)$. The next results show that density of Θ_S in G_S —the topological form of the ergodicity hypothesis—is necessary but not sufficient.

Lemma 4.4. *If $\overline{\sigma(I_S)} \neq G_S$ then $\Xi_\beta(K, S) \neq \{1\}$ for every $\beta > 0$, and the KMS_β state is never unique.*

Proof. Choose $\chi \neq 1$ trivial on the proper closed subgroup $\overline{\sigma(I_S)}$; then every term of (1.1) vanishes. $\square$

Definition 4.5. Let $P_1 = \{\ell \text{ prime} : \ell \equiv 1 \pmod{3}\}$ and let $R = \{r_1 < r_2 < \dots\}$ be an infinite set of primes $\equiv 2 \pmod{3}$ with $r_k > 2^{2^k}$, which exists by Dirichlet's theorem. Put $S = \{3\} \cup P_1 \cup R$.

Lemma 4.6. *For S as in Definition 4.5: $\sum_{\ell \in R} \ell^{-\beta} < \infty$ for every $\beta > 0$; and $\sum_{\ell \in S} \ell^{-1} = \infty$, so $\beta_c(S) = 1$ and $\zeta_S(\beta) = \infty$ for $0 < \beta \leq 1$. Moreover, letting $\chi \in \widehat{G_S}$ be the quadratic character at the component 3, i.e. $\chi(u) = \left(\frac{u_3 \bmod 3}{3}\right)$, one has $\chi(\sigma_3) = 1$, $\chi(\sigma_\ell) = 1$ for $\ell \in P_1$ and $\chi(\sigma_\ell) = -1$ for $\ell \in R$, so that*

$$\sum_{\ell \in S} \ell^{-\beta} |1 - \chi(\sigma_\ell)|^2 = 4 \sum_{\ell \in R} \ell^{-\beta} < \infty$$

and $\chi \in \Xi_\beta(\mathbb{Q}, S)$ for every $\beta > 0$.

Proof. $r_k^{-\beta} < 2^{-\beta 2^k}$. Divergence over P_1 is Dirichlet's theorem in the form $\sum_{\ell \leq x, \ell \equiv 1(3)} \ell^{-1} = \frac{1}{2} \log \log x + O(1)$. For the character: $(\sigma_3)_3 = 1$ and $(\sigma_\ell)_3 = \ell$ for $\ell \neq 3$, so $\chi(\sigma_\ell) = \left(\frac{\ell}{3}\right)$, which is $+1$ for $\ell \equiv 1$ and -1 for $\ell \equiv 2 \pmod{3}$. $\square$

Proposition 4.7. *For S as in Definition 4.5, $\Theta_S = \sigma(I_S)$ is dense in $G_S = \widehat{\mathbb{Z}}_S^\times$.*

Proof. Density is equivalent to surjectivity of $\Theta_S \rightarrow \prod_{\ell \in F} (\mathbb{Z}/\ell^{k_\ell} \mathbb{Z})^\times$ for every finite $F \subset S$ and all exponents $k_\ell \geq 1$. Put $M = \prod_{\ell \in F} \ell^{k_\ell}$ and let $\iota : (\mathbb{Z}/M\mathbb{Z})^\times \xrightarrow{\sim} \prod_{\ell \in F} (\mathbb{Z}/\ell^{k_\ell} \mathbb{Z})^\times$ be the Chinese remainder isomorphism. For $\ell \in S \setminus F$ the image of σ_ℓ is $\iota(\ell \bmod M)$, since $(\sigma_\ell)_{\ell'} = \ell$ for every $\ell' \in F$. So it suffices to prove that $D := \langle \ell \bmod M : \ell \in S \setminus F \rangle = (\mathbb{Z}/M\mathbb{Z})^\times$.

Let $K_M = (\mathbb{Z}/M\mathbb{Z})^\times$ if $3 \nmid M$ and $K_M = \{a : a \equiv 1 \pmod{3}\}$ if $3 \mid M$, a subgroup of index 1 or 2.

Step 1: $D \supseteq K_M$. Let $a \in K_M$. If $3 \mid M$ then $a \equiv 1 \pmod{3}$ and Dirichlet's theorem provides infinitely many primes $\ell \equiv a \pmod{M}$; each lies in $P_1 \subset S$ and all but finitely many avoid F . If $3 \nmid M$, choose $b \in (\mathbb{Z}/3M\mathbb{Z})^\times$ with $b \equiv a \pmod{M}$ and $b \equiv 1 \pmod{3}$, and apply Dirichlet's theorem to the modulus $3M$.

Step 2: $D = (\mathbb{Z}/M\mathbb{Z})^\times$. If $3 \nmid M$, Step 1 suffices. If $3 \mid M$, choose $r \in R \setminus F$, possible since R is infinite; then $r \equiv 2 \pmod{3}$, so $r \bmod M \notin K_M$, while $r \bmod M \in D$. As $[(\mathbb{Z}/M\mathbb{Z})^\times : K_M] = 2$, D is everything. $\square$

Remark 4.8. Step 2 consumed exactly one element of R . Density holds verbatim if R is a single prime, and fails only if $R = \emptyset$, in which case $\Theta_S \subseteq \ker \chi$ and Lemma 4.4 applies. This is the source of the phenomenon below.

Proposition 4.9. *For S as in Definition 4.5 and $0 < \beta \leq 1$,*

$$\Xi_\beta(\mathbb{Q}, S) = \{1, \chi\} \cong \mathbb{Z}/2\mathbb{Z}, \quad \Xi_\beta^\perp = \ker \chi, \quad G_S / \Xi_\beta^\perp \cong \mathbb{Z}/2\mathbb{Z},$$

while $\Xi_\beta = \widehat{G_S}$ for $\beta > 1$.

Proof. The case $\beta > 1$ is Lemma 3.5, since $\zeta_S(\beta) < \infty$. Let $0 < \beta \leq 1$ and $\chi' \in \widehat{G_S}$ be nontrivial, factoring through $(\mathbb{Z}/M\mathbb{Z})^\times$ for a modulus M supported on a finite $F \subset S$, with associated Dirichlet character $\tilde{\chi}'$, so that $\chi'(\sigma_\ell) = \tilde{\chi}'(\ell)$ for $\ell \in S \setminus F$. Since F is finite, membership in Ξ_β depends only on $\tilde{\chi}'$ and is equivalent to $\sum_{\ell \in S, \tilde{\chi}'(\ell) \neq 1} \ell^{-\beta} < \infty$.

Case 1: $\tilde{\chi}'|_{K_M} \neq 1$. Pick $a \in K_M$ with $\tilde{\chi}'(a) \neq 1$. As in Step 1 of Proposition 4.7 there are infinitely many $\ell \in P_1$ with $\ell \equiv a \pmod{M}$, of positive relative density among the primes, so $\sum \ell^{-\beta} \geq \sum \ell^{-1} = \infty$ over that set; hence $\chi' \notin \Xi_\beta$.

Case 2: $\tilde{\chi}'|_{K_M} = 1$. If $3 \nmid M$ then K_M is everything and $\tilde{\chi}' = 1$, so χ' is trivial on all σ_ℓ , $\ell \in S \setminus F$; these generate $(\mathbb{Z}/M\mathbb{Z})^\times$ by Proposition 4.7, so $\chi' = 1$, excluded. If $3 \mid M$ then $\tilde{\chi}'$ factors through the group of order two $(\mathbb{Z}/M\mathbb{Z})^\times/K_M$ and is either trivial—excluded as before—or equal to $\tilde{\chi}$, in which case $\chi'\tilde{\chi}$ has trivial associated character, hence is trivial by the same generation argument, so $\chi' = \chi$.

Finally $\chi \in \Xi_\beta$ by Lemma 4.6, and $\Xi_\beta^\perp = \ker \chi$ has index 2 since χ is onto $\{\pm 1\}$. $\square$

Theorem 4.10. *Let S be as in Definition 4.5 and $0 < \beta \leq 1$. Then:*

- (1) Θ_S is dense in G_S ;
- (2) $\zeta_S(\beta) = \infty$, so there are no Gibbs states and every $\nu \in M_\beta(\mathbb{Q}, S)$ gives G_S measure zero;
- (3) the KMS_β simplex is a segment, affinely isomorphic to $\text{Prob}(\mathbb{Z}/2\mathbb{Z})$, with exactly two extreme points $\varphi_\pm$;
- (4) $\widehat{\mathbb{Z}}_S^\times$ acts transitively on $\{\varphi_+, \varphi_-\}$ through $\widehat{\mathbb{Z}}_S^\times / \ker \chi \cong \mathbb{Z}/2\mathbb{Z}$.

Proof. (1) is Proposition 4.7; (2) is Lemma 4.6 with Theorem 3.6; (3) and (4) follow from Theorem 3.6 and Proposition 4.9. $\square$

The invariant. The two extreme points are separated by an explicit function.

Definition 4.11. For $x \in \widehat{\mathbb{Z}}_S$ with all coordinates nonzero and $\sum_{\ell \in R} v_\ell(x) < \infty$, set

$$(4.1) \quad \Psi(x) = \left(\frac{u_3(x) \bmod 3}{3} \right) \cdot (-1)^{\sum_{\ell \in R} v_\ell(x)} \in \{\pm 1\}, \quad u_3(x) = x_3 3^{-v_3(x)}.$$

Proposition 4.12. (1) Ψ is defined ν -almost everywhere for every $\nu \in M_\beta(\mathbb{Q}, S)$ and every $\beta > 0$; the defining condition holds a.s. precisely because $\sum_{\ell \in R} \ell^{-\beta} < \infty$, by the Borel–Cantelli lemma.

- (2) $\Psi(nx) = \Psi(x)$ for every $n \in \mathbb{N}_S$ and every x in the domain, an identity of functions and not merely an almost-everywhere statement.
- (3) Under the symmetric measure, $\Psi = \pm 1$ each with probability $\frac{1}{2}$; conditioning on $\{\Psi = \pm 1\}$ gives the two extreme points $\varphi_\pm$ of Theorem 4.10, whose measures are mutually singular.

Proof. (1) $\sum_{\ell \in R} \pi_\beta(v_\ell \geq 1) = \sum_{\ell \in R} \ell^{-\beta} < \infty$ by Lemma 4.6, so a.s. only finitely many $\ell \in R$ have $v_\ell \geq 1$; each v_ℓ is finite a.s. by Lemma 2.10(3).

(2) It suffices to treat $n = \ell_0$ prime. Multiplication by ℓ_0 sends $v_\ell \mapsto v_\ell + \delta_{\ell, \ell_0}$ and $u_\ell \mapsto (\theta_{\ell_0})_\ell u_\ell$. If $\ell_0 = 3$: u_3 is unchanged, since $(\theta_3)_3 = 1$, and no v_ℓ with $\ell \in R$ changes. If $\ell_0 \in P_1$: $u_3 \mapsto \ell_0 u_3$ with $\left(\frac{\ell_0}{3}\right) = 1$, and again no v_ℓ , $\ell \in R$, changes. If $\ell_0 \in R$: $u_3 \mapsto \ell_0 u_3$ with $\left(\frac{\ell_0}{3}\right) = -1$, so the first factor changes sign, and $v_{\ell_0} \mapsto v_{\ell_0} + 1$, so the second factor changes sign as well; the product is unchanged.

(3) Under the symmetric measure the conditional law of u given v is Haar, so $\chi(u) = \pm 1$ with probability $\frac{1}{2}$ each, independently of v ; hence so is Ψ . The sets $\{\Psi = \pm 1\}$ are invariant by (2), so the conditioned measures lie in $M_\beta(\mathbb{Q}, S)$; they are mutually singular, and their midpoint is the symmetric measure, which by Theorem 4.10(3) is the midpoint of a segment. A segment has exactly two extreme points. $\square$

Remark 4.13 (Ψ as an order parameter). Ψ is measurable with respect to the tail σ -algebra of the family $(v_\ell)_{\ell \in S}$ up to the single unit coordinate u_3 : changing the valuations at any finite set of primes leaves it unchanged, because a change at $\ell \in R$ flips both factors of (4.1) at once and a change elsewhere flips neither. It is therefore a macroscopic observable in the sense of statistical mechanics—an asymptotically conserved charge that no local operation can alter—and the two extreme points $\varphi_\pm$ are the two phases in which it is sharp. This is the concrete form taken here by the general principle that symmetry breaking is detected by a tail-measurable order parameter; the Kolmogorov zero–one law of Lemma 3.1 is exactly the statement that no such parameter exists when Ξ_β is trivial.

Remark 4.14 (Where density and uniqueness part company). Write $R_\chi = \{\ell \in S : \chi(\sigma_\ell) \neq 1\}$; here $R_\chi = R$. Then

$$\overline{\Theta_S} = G_S \iff R_\chi \neq \emptyset \text{ for every nontrivial } \chi, \quad \chi \notin \Xi_\beta \iff \sum_{\ell \in R_\chi} \ell^{-\beta} = \infty.$$

Density asks for the *existence* of a prime detecting χ ; uniqueness asks for an *abundance* of them, weighted by the same $\ell^{-\beta}$ that defines the dynamics. The invariant Ψ makes the gap concrete: it is built from the almost surely finitely many indices $\ell \in R$ at which x is divisible by ℓ , and such a finite count exists exactly when the Borel–Cantelli series converges. Were $\sum_{\ell \in R} \ell^{-\beta} = \infty$, the exponent in (4.1) would be infinite almost surely and Ψ would be undefined on a conull set.

Remark 4.15 (Kakutani dichotomy). The criterion (1.1) is of Kakutani type: by Lemma 3.2 and Proposition 3.3 the obstruction is the convergence of $\prod_{\mathfrak{p} \in S} |m_{\mathfrak{p}}|$, and this product is nonzero exactly when $\sum_{\mathfrak{p}} (1 - |m_{\mathfrak{p}}|) < \infty$, which is the shape of Kakutani’s theorem on equivalence versus mutual singularity of infinite product measures. The dichotomy is between the translates of ν being pairwise equivalent (ergodicity, uniqueness) and a nontrivial invariant splitting existing (mutual singularity, symmetry breaking), and it is decided by a convergent-or-divergent series. At the level of general principle this is the familiar fact that a minimal action need not be uniquely ergodic—already visible for flows on surfaces [3]—and no novelty is claimed for the principle; what is new is that the exact place where the two conditions separate can be written down, and that it is a condition on a Dirichlet series attached to S .

Cascades. By Lemma 3.5 the family $\{\Xi_\beta\}_{\beta > 0}$ is a nested increasing chain of subgroups of $\widehat{G_S}$, equal to $\{1\}$ for small β in the uniqueness range and to $\widehat{G_S}$ for $\beta > \beta_c(S)$. Each jump is a phase transition. The classical systems have the single jump $\{1\} \rightarrow \widehat{G_S}$ at $\beta = 1$; intermediate jumps can be prescribed.

Proposition 4.16. *Let $0 < \beta_2 < \beta_1 < 1$. There is a set of primes S with $\beta_c(S) = 1$, Θ_S dense in G_S , and*

$$\Xi_\beta(\mathbb{Q}, S) = \begin{cases} \{1\}, & 0 < \beta \leq \beta_2, \\ \{1, \chi_5\}, & \beta_2 < \beta \leq \beta_1, \\ \{1, \chi_3, \chi_5, \chi_3\chi_5\}, & \beta_1 < \beta \leq 1, \\ \widehat{G_S}, & \beta > 1, \end{cases}$$

where $\chi_3, \chi_5 \in \widehat{G_S}$ are the quadratic characters at the components 3 and 5. Thus $\mathcal{A}_{\mathbb{Q}, S}$ has exactly three phase transitions, at β_2, β_1 and 1, with residual symmetry groups

$$G_S \supsetneq \ker \chi_5 \supsetneq \ker \chi_3 \cap \ker \chi_5 \supsetneq \{1\}.$$

Proof. The set. Work modulo 15 and note $(\mathbb{Z}/15\mathbb{Z})^\times = \{1, 2, 4, 7, 8, 11, 13, 14\}$ is isomorphic to $(\mathbb{Z}/3\mathbb{Z})^\times \times (\mathbb{Z}/5\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$. Define

$$S = \{3, 5\} \cup P \cup R_1 \cup R_2,$$

where

$$P = \{\ell : \ell \equiv 1 \text{ or } 4 \pmod{15}\},$$

and R_1, R_2 are infinite sets of primes with

$$R_1 \subseteq \{\ell \equiv 11 \pmod{15}\}, \quad R_2 \subseteq \{\ell \equiv 7 \pmod{15}\},$$

chosen so that $|R_i \cap [1, x]| \asymp x^{\beta_i}$. Such sets exist: by Dirichlet's theorem each of the classes 11 and 7 contains infinitely many primes, and one selects greedily one prime of the class in each interval $[m^{1/\beta_i}, 2m^{1/\beta_i}]$, $m = 1, 2, \dots$. The classes $\{1, 4\}$, $\{11\}$, $\{7\}$ are pairwise disjoint, so the union is disjoint. Note $\ell \equiv 11$ means $\ell \equiv 2 \pmod{3}$ and $\ell \equiv 1 \pmod{5}$, while $\ell \equiv 7$ means $\ell \equiv 1 \pmod{3}$ and $\ell \equiv 2 \pmod{5}$.

By Abel summation, for either i ,

$$(4.2) \quad \sum_{\ell \in R_i} \ell^{-\beta} \asymp \int_2^\infty x^{-\beta} d(x^{\beta_i}) \asymp \int_2^\infty x^{\beta_i - \beta - 1} dx,$$

which converges for $\beta > \beta_i$ and diverges for $\beta \leq \beta_i$, the case $\beta = \beta_i$ giving $\int dx/x$. Since P has positive natural density among the primes, $\sum_{\ell \in P} \ell^{-1} = \infty$, so $\beta_c(S) = 1$ and $\zeta_S(\beta) = \infty$ for $\beta \leq 1$, while $\zeta_S(\beta) < \infty$ for $\beta > 1$; the last case of the displayed formula is then Lemma 3.5.

The subgroup K_M . Let M be a modulus supported on S and set

$$K_M = \{a \in (\mathbb{Z}/M\mathbb{Z})^\times : a \bmod \gcd(M, 15) \in \langle 4 \rangle\},$$

with the convention that the condition is vacuous when $\gcd(M, 15) = 1$. Since $4^2 = 16 \equiv 1 \pmod{15}$, the set $\langle 4 \rangle = \{1, 4\}$ is a subgroup of $(\mathbb{Z}/15\mathbb{Z})^\times$ of order 2, so K_M is a subgroup of index 1, 2, 2 or 4 according as $\gcd(M, 15)$ is 1, 3, 5 or 15. By the Chinese remainder theorem and Dirichlet's theorem, K_M is exactly the set of classes modulo M realized by primes of P outside any prescribed finite set: given $a \in K_M$, choose by CRT a class b modulo $\text{lcm}(M, 15)$ with $b \equiv a \pmod{M}$ and $b \in \{1, 4\} \pmod{15}$, and apply Dirichlet's theorem to the modulus $\text{lcm}(M, 15)$.

Characters trivial on K_M . Under $(\mathbb{Z}/15\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ the element 4 corresponds to $(0, 2)$, being the square of a primitive root modulo 5; hence

$$(\mathbb{Z}/15\mathbb{Z})^\times / \langle 4 \rangle \cong \mathbb{Z}/2\mathbb{Z} \times (\mathbb{Z}/4\mathbb{Z} / \langle 2 \rangle) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z},$$

and the characters of $(\mathbb{Z}/M\mathbb{Z})^\times$ trivial on K_M are precisely the four characters $1, \tilde{\chi}_3, \tilde{\chi}_5, \tilde{\chi}_3\tilde{\chi}_5$, where $\tilde{\chi}_3$ is the quadratic character modulo 3 and $\tilde{\chi}_5$ the quadratic character modulo 5, inflated to modulus M . In particular a quartic character $\tilde{\psi}_5$ modulo 5 is *not* trivial on K_M : $\tilde{\psi}_5(4) = \tilde{\psi}_5(2)^2 = (\pm i)^2 = -1$.

Density of Θ_S . As in Proposition 4.7 it suffices to show that $D_M := \langle \ell \bmod M : \ell \in S \setminus F \rangle$ equals $(\mathbb{Z}/M\mathbb{Z})^\times$ for every modulus M supported on S and every finite $F \subset S$. By the previous paragraph $D_M \supseteq K_M$. If $\gcd(M, 15) = 1$ we are done. Otherwise pick $\ell_1 \in R_1 \setminus F$ and $\ell_2 \in R_2 \setminus F$, possible since R_1, R_2 are infinite. Their images in $(\mathbb{Z}/M\mathbb{Z})^\times / K_M \hookrightarrow (\mathbb{Z}/15\mathbb{Z})^\times / \langle 4 \rangle$ are the classes of 11 and 7, which are distinct and nontrivial: indeed $\tilde{\chi}_3(11) = -1$, $\tilde{\chi}_5(11) = 1$, $\tilde{\chi}_3(7) = 1$, $\tilde{\chi}_5(7) = -1$, since 2 is a non-residue modulo 5. Two distinct nontrivial elements generate $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, so $D_M = (\mathbb{Z}/M\mathbb{Z})^\times$.

Computation of Ξ_β . Let $0 < \beta \leq 1$ and let $\chi' \in \widehat{G_S}$ be nontrivial, factoring through $(\mathbb{Z}/M\mathbb{Z})^\times$ for a modulus M supported on a finite $F \subset S$, with associated Dirichlet character $\tilde{\chi}'$, so that $\chi'(\sigma_\ell) = \tilde{\chi}'(\ell)$ for $\ell \in S \setminus F$. As F is finite, membership of χ' in Ξ_β depends only on $\tilde{\chi}'$ and is equivalent to $\sum_{\ell \in S, \tilde{\chi}'(\ell) \neq 1} \ell^{-\beta} < \infty$.

If $\tilde{\chi}'|_{K_M} \neq 1$, choose $a \in K_M$ with $\tilde{\chi}'(a) \neq 1$; by the description of K_M there are infinitely many $\ell \in P$ with $\ell \equiv a \pmod{M}$, forming a set of positive relative density among the primes, so $\sum \ell^{-\beta} \geq \sum \ell^{-1} = \infty$ over that set and $\chi' \notin \Xi_\beta$. This disposes in particular of every character whose 5-part is quartic.

If $\tilde{\chi}'|_{K_M} = 1$ then $\tilde{\chi}' \in \{1, \tilde{\chi}_3, \tilde{\chi}_5, \tilde{\chi}_3\tilde{\chi}_5\}$. If $\tilde{\chi}' = 1$ then χ' is trivial on σ_ℓ for every $\ell \in S \setminus F$, and these generate by the density just proved, so $\chi' = 1$, contrary to assumption; the same argument shows that $\tilde{\chi}' = \tilde{\chi}_3$ forces $\chi' = \chi_3$, and likewise for $\tilde{\chi}_5$ and $\tilde{\chi}_3\tilde{\chi}_5$. It remains to evaluate the three sums, using that $\tilde{\chi}_3$ and $\tilde{\chi}_5$ are trivial on P :

$$\tilde{\chi}' = \tilde{\chi}_3 : \quad \{\ell \in S : \tilde{\chi}_3(\ell) \neq 1\} = R_1 \cup (\text{finite}),$$

$$\tilde{\chi}' = \tilde{\chi}_5 : \quad \{\ell \in S : \tilde{\chi}_5(\ell) \neq 1\} = R_2 \cup (\text{finite}),$$

$$\tilde{\chi}' = \tilde{\chi}_3\tilde{\chi}_5 : \quad \{\ell \in S : \tilde{\chi}_3\tilde{\chi}_5(\ell) \neq 1\} = R_1 \cup R_2 \cup (\text{finite}),$$

because $R_1 \equiv 2 \pmod{3}$ and $\equiv 1 \pmod{5}$, while $R_2 \equiv 1 \pmod{3}$ and $\equiv 2 \pmod{5}$, so on R_1 the pair $(\tilde{\chi}_3, \tilde{\chi}_5)$ takes the value $(-1, 1)$ and on R_2 the value $(1, -1)$; the finite sets account for $\ell \in \{3, 5\} \cup F$. By (4.2) the first sum converges iff $\beta > \beta_1$, the second iff $\beta > \beta_2$, and the third iff $\beta > \beta_1$, using $\beta_2 < \beta_1$. Hence

$$\Xi_\beta \cap \{\chi_3, \chi_5, \chi_3\chi_5\} = \begin{cases} \emptyset, & \beta \leq \beta_2, \\ \{\chi_5\}, & \beta_2 < \beta \leq \beta_1, \\ \{\chi_3, \chi_5, \chi_3\chi_5\}, & \beta_1 < \beta \leq 1, \end{cases}$$

which together with the exclusion of all other characters gives the stated formula. The residual symmetry groups are $\Xi_\beta^\perp$, and $\{\chi_3, \chi_5\}$ being independent quadratic characters, these are G_S , $\ker \chi_5$, $\ker \chi_3 \cap \ker \chi_5$ and $\{1\}$ respectively. $\square$

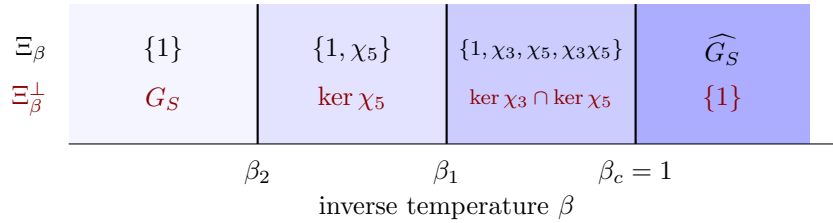


FIGURE 2. The phase diagram of Proposition 4.16. Upper row: the obstruction group Ξ_β , increasing with β by Lemma 3.5. Lower row: the residual symmetry group $\Xi_\beta^\perp$, decreasing along the chain. The classical systems have the single jump $\{1\} \rightarrow \widehat{G_S}$ at $\beta = 1$.

4.3. The unit group and Leopoldt's conjecture. The units enter through the left-hand term of (2.1), and their effect is opposite to that of the class group.

Proposition 4.17. *Enlarging $\overline{\mathcal{O}_{K,+}^\times}$ shrinks G_S , hence $\widehat{G_S}$; all else equal, this makes the condition $\Xi_\beta(K, S) = \{1\}$ easier to satisfy. Moreover, let p be a rational prime, suppose S contains every prime of K above p , and let r be the $\mathbb{Z}$ -rank of $\mathcal{O}_{K,+}^\times$. Then the closure of $\mathcal{O}_{K,+}^\times$ in $\prod_{\mathfrak{p}|p} \mathcal{O}_{\mathfrak{p}}^\times$ is a $\mathbb{Z}_p$ -module of rank at most r .*

Proof. The first statement is immediate from (2.1) and duality: characters of G_S are characters of $\mathbb{I}_S$ trivial on $\overline{K_{S,+}^\times}$, a stronger condition when the closure is larger. For the second, the closure of a finitely generated abelian group of rank r inside a p -adic Lie group is a $\mathbb{Z}_p$ -module generated by r elements, hence of $\mathbb{Z}_p$ -rank at most r . $\square$

Remark 4.18. Equality of the rank with r in Proposition 4.17 is precisely the non-vanishing of the p -adic regulator, i.e. Leopoldt's conjecture for K at p , which is known for K abelian over $\mathbb{Q}$ (Ax-Brumer) and open in general. We state only the unconditional inequality, and we emphasize that Theorem 3.9 is unaffected: the criterion is phrased through splitting behaviour in finite abelian

extensions and does not refer to how the local units assemble. Leopoldt's conjecture governs the description of G_S as a topological group, not the classification of its KMS states.

4.4. Summary of the comparison.

	$\mathbb{Q}$	general K
parameter group	$\widehat{\mathbb{Z}}_S^\times$	$G_S, 1 \rightarrow U_S/\mathcal{O}_{K,+}^\times \rightarrow G_S \rightarrow \text{Cl}_S^+ \rightarrow 1$
coordinates	θ_ℓ elementary	ideles; no canonical generator
obstruction test	Dirichlet character	$\chi(\text{Frob}_{\mathfrak{p}})$, ray class character
uniqueness for $S = P_K$	Dirichlet	Chebotarev, i.e. $L(1, \chi) \neq 0$
unramified obstructions	none ($h_{\mathbb{Q}} = 1$)	subfields of H^+ when $h_K^+ > 1$
units	trivial	shrink G_S , aid uniqueness

5. SPARSE SETS: THE CLOSED LOW-TEMPERATURE PHASE AND SOPHIE GERMAIN PRIMES

We return to $K = \mathbb{Q}$ and apply the theory to sets of primes that are sparse in the sense of sieve theory. The results of this section concern the location and nature of the transition rather than the residual symmetry, and they show that the phase diagram of a localized system can differ from that of [1] even when no obstruction group is present.

Definition 5.1. The *critical exponent* of a set S of primes is

$$\beta_c(S) = \inf \left\{ \beta > 0 : \sum_{\ell \in S} \ell^{-\beta} < \infty \right\} \in [0, 1],$$

with $\beta_c(S) = 0$ if S is finite. We call S *Brun-summable* if $\sum_{\ell \in S} \ell^{-1} < \infty$.

Theorem 5.2 (Location and nature of the transition). *Let S be a set of primes and $\beta_c = \beta_c(S)$.*

- (1) *For $\beta > \beta_c$ the simplex is $\text{Prob}(G_S)$, with extreme points of type I_∞ permuted freely and transitively by G_S (Theorem 3.6(2)).*
- (2) *For $0 < \beta < \beta_c$ there are no Gibbs states, and the simplex is $\text{Prob}(G_S/\Xi_\beta^\perp)$, a point exactly when the criterion of Theorem 3.9 holds.*
- (3) *$\mathcal{A}_{\mathbb{Q},S}$ has a transition, i.e. both regimes are nonempty, if and only if $\beta_c > 0$.*
- (4) *If $\beta_c > 0$ and $\sum_{\ell \in S} \ell^{-\beta_c} < \infty$, then the type I_∞ phase is the closed half-line $[\beta_c, \infty)$ and*

$$\lim_{\beta \downarrow \beta_c} \zeta_S(\beta) = \zeta_S(\beta_c) < \infty,$$

so the partition function does not diverge at the transition. This hypothesis holds whenever S is Brun-summable with $\beta_c = 1$, since then $\sum_{\ell} \ell^{-\beta_c} = \sum_{\ell} \ell^{-1} < \infty$.

Proof. (1)–(3) are Theorem 3.6 and Definition 5.1. For (4), if $\sum_{\ell} \ell^{-\beta_c} < \infty$ then $\zeta_S(\beta_c) < \infty$ and Theorem 3.6(2) applies at $\beta = \beta_c$, so the endpoint lies in the type I_∞ phase. As $\beta \downarrow \beta_c$ one has $\ell^{-\beta} \uparrow \ell^{-\beta_c}$ for each ℓ , so by monotone convergence $\sum_{\ell} \ell^{-\beta} \rightarrow \sum_{\ell} \ell^{-\beta_c} < \infty$ and the Euler products converge to $\zeta_S(\beta_c)$. The last sentence is the definition of Brun-summability together with $\beta_c = 1$. $\square$

Remark 5.3. Item (4) is sharp and marks a genuine difference from the classical system. For S the set of all primes, $\beta_c = 1$ but S is not Brun-summable: $\sum_{\ell} \ell^{-1} = \infty$ by Mertens, $\zeta(\beta_c) = \infty$, and $\zeta(\beta) \rightarrow \infty$ as $\beta \downarrow 1$ through the pole. Thus:

	β_c	$\zeta_S(\beta_c)$	phase of β_c
all primes (Bost–Connes)	1	∞	high-temperature
Brun-summable with $\beta_c = 1$	1	$< \infty$	type I_∞

The two systems have the same critical temperature and opposite behaviour at it; the critical point belongs to the symmetric phase in one case and to the symmetry-broken phase in the other.

The distinction is invisible in the exponent: it is the difference between Mertens’ theorem and Brun’s theorem.

Remark 5.4. For $0 < \beta_c < 1$ the value $\zeta_S(\beta_c)$ is not determined by β_c alone; both convergence and divergence occur among sets with the same convergence exponent, as one sees by comparing $\ell_k \approx \exp(k^{1/\beta_c})$ with $\ell_k \approx \exp(k^{1/\beta_c}/\log k)$. The Brun-summable case with $\beta_c = 1$ is the one relevant below.

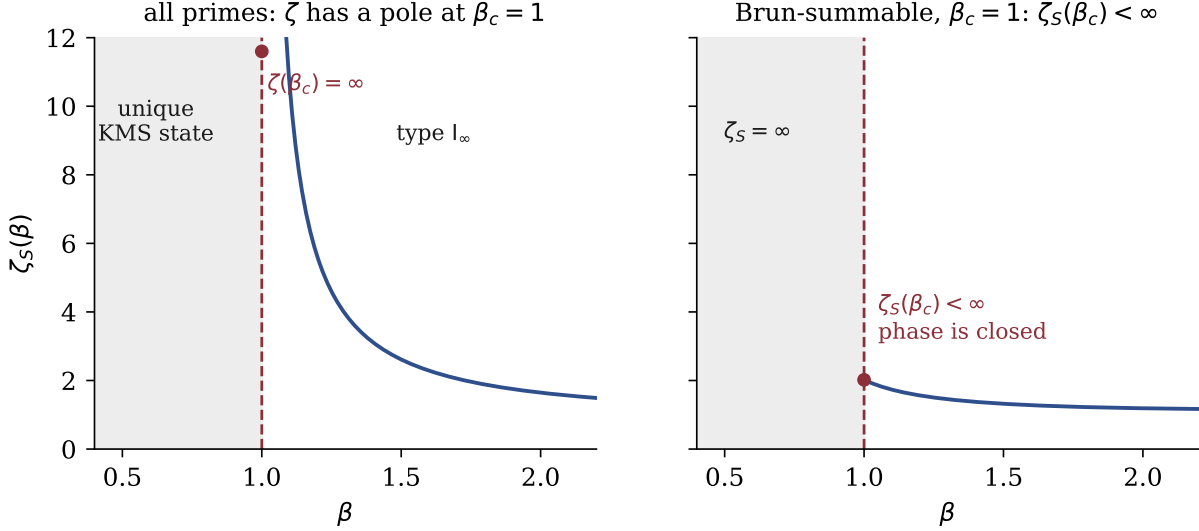


FIGURE 3. Remark 5.3: the same critical temperature, opposite behaviour at it. Left, the classical system, where ζ has a pole at $\beta_c = 1$ and the critical point lies in the high-temperature phase. Right, a Brun-summable set with $\beta_c = 1$, where $\zeta_S(\beta_c) < \infty$ and the type I_∞ phase is closed.

5.1. Sophie Germain primes. Let $S\mathcal{G} = \{p \text{ prime} : 2p + 1 \text{ prime}\}$, $\pi_{S\mathcal{G}}(x) = |S\mathcal{G} \cap [1, x]|$, and $S_\infty = \{2\} \cup S\mathcal{G}$; by Proposition 2.7, $\mathcal{A}_{\mathbb{Q}, S_\infty} = \varinjlim_k \mathcal{A}_{\mathbb{Q}, T_k}$ with $T_k = \{2, p_1, \dots, p_k\}$, and $\mathcal{A}_{\mathbb{Q}, S_\infty} \subseteq \mathcal{A}_{BC}$.

Theorem 5.5. *Let $\beta_c = \beta_c(S_\infty)$.*

- (1) (Unconditional.) $\pi_{S\mathcal{G}}(x) \ll x/\log^2 x$ [9]; hence S_∞ is Brun-summable, $\zeta_{S_\infty}(\beta) < \infty$ for all $\beta \geq 1$, and $\beta_c \leq 1$.
- (2) If $\pi_{S\mathcal{G}}(x) \gg x/\log^A x$ for some $A > 0$ —in particular under the Hardy–Littlewood conjecture $\pi_{S\mathcal{G}}(x) \sim 2C_2 x/\log^2 x$ [8]—then $\beta_c = 1$.
- (3) Under the hypothesis of (2), ζ_{S_∞} is finite on $[1, \infty)$ and identically $+\infty$ on $(0, 1)$, and the type I_∞ phase of $\mathcal{A}_{\mathbb{Q}, S_\infty}$ is exactly $[1, \infty)$: a closed low-temperature phase whose critical point is $\beta_c = 1$.
- (4) $\mathcal{A}_{\mathbb{Q}, S_\infty}$ has a phase transition if and only if $S\mathcal{G}$ has positive convergence exponent; in particular a phase transition implies that $S\mathcal{G}$ is infinite.
- (5) For β in the range where $\zeta_{S_\infty}(\beta) = \infty$, the KMS_β state is unique if and only if, for every modulus M supported on S_∞ and every proper subgroup $H < (\mathbb{Z}/M\mathbb{Z})^\times$,

$$\sum_{\substack{p \in S\mathcal{G} \\ p \bmod M \notin H}} p^{-\beta} = \infty;$$

this holds under the Hardy–Littlewood conjecture with congruence conditions.

Proof. (1) Abel summation: with $\pi_{\mathcal{SG}}(t) \leq Ct/\log^2 t$ for $t \geq t_0$,

$$\sum_{p \in \mathcal{SG}, p \leq x} \frac{1}{p} = \frac{\pi_{\mathcal{SG}}(x)}{x} + \int_{t_0}^x \frac{\pi_{\mathcal{SG}}(t)}{t^2} dt + O(1) \leq O(1) + C \int_{t_0}^x \frac{dt}{t \log^2 t} = O(1).$$

Hence $\sum_{p \in \mathcal{SG}} p^{-1} < \infty$, $\beta_c \leq 1$, and $\zeta_{S_\infty}(\beta) < \infty$ for $\beta \geq 1$ since $\ell^{-\beta} \leq \ell^{-1}$.

(2) For $0 < \beta < 1$, Abel summation with the lower bound gives

$$\sum_{p \in \mathcal{SG}, p \leq x} p^{-\beta} \geq \beta \int_{t_0}^x \frac{\pi_{\mathcal{SG}}(t)}{t^{\beta+1}} dt \gg \int_{t_0}^x \frac{dt}{t^\beta \log^A t} \gg \frac{x^{1-\beta}}{\log^A x} \rightarrow \infty,$$

so $\beta_c \geq 1$; with (1), $\beta_c = 1$.

(3) Combine (1), (2) and Theorem 5.2(4).

(4) Theorem 5.2(3).

(5) The first assertion is Theorem 3.9 specialized to $K = \mathbb{Q}$, where the abelian extensions unramified outside $S \cup \infty$ are the subfields of cyclotomic fields $\mathbb{Q}(\zeta_M)$ with M supported on S , and Frob_p is the class of p modulo M . For the conditional statement, assume that for each admissible residue a one has $|\{p \leq x : p \in \mathcal{SG}, p \equiv a \pmod{M}\}| \gg_M x/\log^2 x$, where

$$A_M = \{a \in (\mathbb{Z}/M\mathbb{Z})^\times : 2a + 1 \in (\mathbb{Z}/M\mathbb{Z})^\times\}$$

is the set of admissible classes. By Lemma 5.6 below, $\langle A_M \rangle = (\mathbb{Z}/M\mathbb{Z})^\times$, so for any proper subgroup H there is $a \in A_M \setminus H$; the primes of $\mathcal{SG}$ in that class then give a divergent sum by the computation in (2), and they all lie outside H . $\square$

Lemma 5.6. $\langle A_M \rangle = (\mathbb{Z}/M\mathbb{Z})^\times$ for every $M \geq 1$.

Proof. Let χ be a character of $(\mathbb{Z}/M\mathbb{Z})^\times$ trivial on A_M ; we show $\chi = 1$. By the Chinese remainder theorem, $\chi = \prod_\ell \chi_\ell$ and $A_M = \prod_\ell A_{\ell^{k_\ell}}$, where $A_{2^k} = (\mathbb{Z}/2^k)^\times$ and, for ℓ odd, $A_{\ell^k} = \{a \in (\mathbb{Z}/\ell^k)^\times : a \not\equiv -2^{-1} \pmod{\ell}\}$. As A_M is a product, each χ_ℓ is constant on $A_{\ell^{k_\ell}}$, say with value c_ℓ , and $\prod_\ell c_\ell = 1$.

For $\ell = 2$: A_{2^k} is the whole group, so $\chi_2 = 1$ and $c_2 = 1$. For $\ell \geq 5$: $|A_{\ell^k}| = \varphi(\ell^k)(1 - \frac{1}{\ell-1}) > \frac{1}{2}\varphi(\ell^k)$, so $A_{\ell^k} A_{\ell^k}^{-1} = (\mathbb{Z}/\ell^k)^\times$, and χ_ℓ , being constant on A_{ℓ^k} , is trivial, so $c_\ell = 1$. For $\ell = 3$: $A_{3^k} = \{a : a \not\equiv 1 \pmod{3}\}$ is the nontrivial coset of the index-two subgroup $K = \{a \equiv 1 \pmod{3}\}$, so $A_{3^k} A_{3^k}^{-1} = K$ and χ_3 is trivial on K ; hence χ_3 is trivial or the quadratic character with kernel K , and $c_3 = \pm 1$ accordingly. Since all other $c_\ell = 1$ and $\prod c_\ell = 1$, we get $c_3 = 1$, so χ_3 is trivial on $A_{3^k} \cup K$, which is everything. Therefore $\chi = 1$. $\square$

Remark 5.7. Theorem 5.5(4) is an equivalence, not an implication in a useful direction. Establishing the transition requires exactly the arithmetic input that is unavailable, a lower bound on the density of $\mathcal{SG}$; and the transition, once granted, yields nothing about $\mathcal{SG}$ beyond the density statement assumed. The construction is a dictionary entry, not a method. The same applies verbatim to the twin primes, to the primes p with $p+2k$ prime for fixed k , and to any set for which a Brun-type upper bound is known and a matching lower bound conjectured: all such systems have the same phase diagram, and none of them distinguishes its defining condition from any other of the same density. What Theorem 5.5 does record is that the two open points—the location of β_c and the uniqueness below it—are governed by the same input, the second in a form refined by congruence conditions.

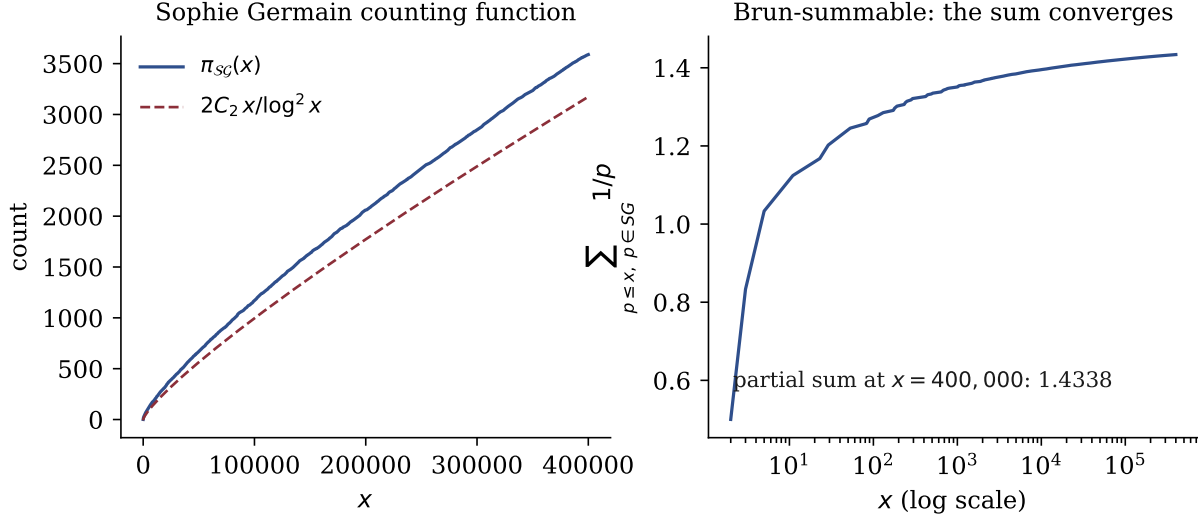


FIGURE 4. Data for Theorem 5.5(1). Left, $\pi_{SG}(x)$ against $2C_2 x / \log^2 x$. Right, the partial sums of $\sum_{p \in SG} 1/p$, whose convergence is the Brun-summability used throughout §5.

5.2. Restriction of the classical phases. Finally we record how the phases of the full system restrict, which explains in what sense the localized systems are cross-sections of the classical one.

Proposition 5.8. *Let S be a set of primes with complement S^c , let $\beta > 1$ and $\chi \in \widehat{\mathbb{Z}}^\times$, and let $\varphi_{\beta, \chi}^{BC}$ be the corresponding extremal KMS_β state of $\mathcal{A}_{BC}$. Then, under the embedding of Proposition 2.7,*

$$\varphi_{\beta, \chi}^{BC}|_{\mathcal{A}_{\mathbb{Q}, S}} = \frac{1}{\zeta_{S^c}(\beta)} \sum_{m \in \mathbb{N}_{S^c}} m^{-\beta} \varphi_{\beta, \text{pr}_S(m)\chi_S}^S, \quad \chi_S = \text{pr}_S(\chi).$$

The map $\chi_S \mapsto \varphi_{\beta, \chi}^{BC}|_{\mathcal{A}_{\mathbb{Q}, S}}$ is injective and $\widehat{\mathbb{Z}}_S^\times$ -equivariant, but if S^c is infinite its image consists entirely of non-extremal states, whose measures have infinitely many atoms. For $0 < \beta \leq 1$ the unique KMS_β state of $\mathcal{A}_{BC}$ restricts to the symmetric state of $\mathcal{A}_{\mathbb{Q}, S}$.

Proof. Decompose $\ell^2(\mathbb{N}^\times) = \bigoplus_{m \in \mathbb{N}_{S^c}} \ell^2(m\mathbb{N}_S)$. Each summand is invariant under $\pi_\chi(\mathcal{A}_{\mathbb{Q}, S})$, and $\epsilon_{mn} \mapsto \epsilon_n$ intertwines $\pi_\chi|_{\mathcal{A}_{\mathbb{Q}, S}}$ on $\ell^2(m\mathbb{N}_S)$ with $\pi_{\text{pr}_S(m)\chi_S}$, while $e^{-\beta H}$ restricts to $m^{-\beta} e^{-\beta H_S}$. Summing and using $\zeta = \zeta_S \zeta_{S^c}$ gives the formula. The atoms $\text{pr}_S(m)\chi_S$ are pairwise distinct, since $\text{pr}_S(m) = \text{pr}_S(m')$ forces $m \equiv m'$ modulo ℓ^k for all $\ell \in S$ and all k , hence $m = m'$; the atom of largest mass is the one with $m = 1$, so χ_S is recoverable, and equivariance follows from $\pi_\chi \circ \theta_v = \pi_{v^{-1}\chi}$. Extremal states correspond to point masses by Theorem 3.6(2), whereas these measures have infinitely many atoms when $\mathbb{N}_{S^c}$ is infinite. The last claim follows from $\widehat{\mathbb{Z}}^\times$ -invariance of the unique state. $\square$

Remark 5.9. So the parameter space of the low-temperature phase projects equivariantly, $\widehat{\mathbb{Z}}^\times \rightarrow \widehat{\mathbb{Z}}_S^\times$, and this is a genuine reduction of the Galois symmetry; what does not descend is extremality, because the degrees of freedom at the primes outside S have been traced out rather than fixed. In particular the transition of $\mathcal{A}_{\mathbb{Q}, S}$ at $\beta_c(S)$ is not the restriction of the transition of $\mathcal{A}_{BC}$ at $\beta = 1$: for Brun-summable S with $\beta_c(S) = 1$ the two occur at the same temperature but on opposite sides, by Remark 5.3.

6. ASSESSMENT AND OPEN PROBLEMS

We separate what is established from what is not.

Established unconditionally. Theorem 3.6 classifies the KMS_β states of $\mathcal{A}_{K,S}$ for every number field K , every set S of finite primes and every $\beta > 0$; Theorem 3.9 converts it into the splitting criterion; Corollary 3.10 recovers the classical theorems of [1, 11] from Chebotarev; Example 3.13, Theorem 4.10 and Example 4.2 exhibit failures of uniqueness with divergent partition function, the last two with dense $\sigma(I_S)$; Proposition 4.12 supplies the invariant; Theorem 5.2 locates the transition and shows the low-temperature phase can be closed; Theorem 5.5(1) is unconditional.

Conditional or incomplete. Theorem 5.5(2),(3),(5) depend on a lower bound for π_{SG} of Hardy–Littlewood strength, in (5) refined by congruence conditions; Remark 4.18 records that the description of G_S as a topological group, though not the criterion, is entangled with Leopoldt’s conjecture.

Relation to the literature. Sections 2 and 3 are, for $S = P_K$, a reformulation of results in [1, 12, 11, 14, 13]; the reduction to measures (Proposition 2.8) uses [10] and [14] directly. It is worth stating precisely where the present work goes beyond [14]. That paper classifies KMS states on C^* -algebras of *non-principal* groupoids, and its content lies in the field of states on the isotropy group algebras. Proposition 2.8 shows that our groupoid is principal on a conull set—the I_S -action is free because the valuation vector is translated freely—so the isotropy data is empty and [14] contributes only the reduction to quasi-invariant measures. Everything after that point is independent of it: the forced product structure of the valuation marginal (Lemma 2.10(3)), the cohomological equation (2.3), and the extraction of the arithmetic obstruction group Ξ_β from a Kakutani product criterion (Propositions 3.3 and 3.4) are what produce Theorem 3.6, and we have found no counterpart to them in [14]. A literature check remains advisable, but the boundary is clear enough that we do not expect a conflict.

The type of the intermediate phases. We record what the ratio-set argument does and does not give.

Proposition 6.1. *Let $\beta > 0$ with $\zeta_{K,S}(\beta) = \infty$, let φ be an extremal KMS_β state of $\mathcal{A}_{K,S}$, and let $M = \pi_\varphi(\mathcal{A}_{K,S})''$. Put*

$$\Gamma_\beta = \langle N\mathfrak{p}^{-\beta} : \mathfrak{p} \in S \rangle \leq \mathbb{R}_{>0}.$$

Then M is a factor of type III, and its Connes invariant satisfies

$$S(M) \subseteq \overline{\Gamma_\beta} \cup \{0\}.$$

Consequently, if the numbers $\log N\mathfrak{p}$, $\mathfrak{p} \in S$, are pairwise commensurable and generate $\log \lambda_0 \cdot \mathbb{Z}$, then $S(M) \subseteq \{0\} \cup \lambda_0^{-\beta\mathbb{Z}}$, so M is of type $\text{III}_{\lambda_0^{-n\beta}}$ for some integer $n \geq 1$ or of type III_0 .

Proof. Extreme points of a KMS simplex give factor states [2, §5.3]. By Proposition 2.8 the groupoid is principal on a conull set, so M is the Krieger factor of the orbit equivalence relation of I_S acting on (Y_S, ν) , and by Krieger’s theorem [6] $S(M)$ is the ratio set of that relation. The Radon–Nikodym cocycle is $\mathfrak{a}\mathfrak{b}^{-1} \mapsto (N\mathfrak{a}/N\mathfrak{b})^{-\beta}$, whose values lie in Γ_β ; the ratio set is contained in the closure of the essential range of the cocycle, hence in $\overline{\Gamma_\beta} \cup \{0\}$.

M is not semifinite. By [6] this means that the relation carries no σ -finite invariant measure equivalent to ν , i.e. that there is no measurable $f : Y_S \rightarrow (0, \infty)$ with $f(\mathfrak{a} \cdot y) = N\mathfrak{a}^\beta f(y)$ for ν -a.e. y and all $\mathfrak{a} \in J_S$. Suppose f exists and put $X = \log f$. In the coordinates of Lemma 2.10, the function $\Phi(v, t) := \int_{G_S} e^{itX(v, g)} d\rho_v(g)$ satisfies, by (2.3) and Lemma 2.10(2), $\Phi(v + a, t) = N\mathfrak{a}^{i\beta t} \Phi(v, t)$ for π_β -a.e. v : this is (3.1) with $\overline{\chi(\sigma_{\mathfrak{a}})}$ replaced by the unimodular constants $c_{\mathfrak{a}} = N\mathfrak{a}^{i\beta t}$, and the proofs of Lemma 3.1 and Proposition 3.3 use nothing else. Fix t , write $c_{\mathfrak{p}} = N\mathfrak{p}^{i\beta t}$ and $m_{\mathfrak{p}}(t) = (1 - t_{\mathfrak{p}})/(1 - c_{\mathfrak{p}}t_{\mathfrak{p}})$. As in the proof of Proposition 3.3, for every finite $F \subset S$ we have $\Phi = P_F G_F$ with $P_F = \prod_{\mathfrak{p} \in F} c_{\mathfrak{p}}^{v_{\mathfrak{p}}}$ and G_F measurable with respect to $\sigma(v_{\mathfrak{p}} : \mathfrak{p} \notin F)$, hence

independent of P_F and bounded by 1. Therefore

$$\psi(t) := \int_{Y_S} e^{itX} d\nu = \mathbb{E}_{\pi_\beta}[\Phi(\cdot, t)] = \mathbb{E}[P_F] \mathbb{E}[G_F], \quad |\psi(t)| \leq \prod_{\mathfrak{p} \in F} |m_{\mathfrak{p}}(t)|,$$

and letting $F \uparrow S$ and using Lemma 3.2 in the form $1 - |m_{\mathfrak{p}}| \geq \frac{1}{2}(1 - |m_{\mathfrak{p}}|^2) \geq \frac{1}{8}t_{\mathfrak{p}}|1 - c_{\mathfrak{p}}|^2$,

$$|\psi(t)| \leq \prod_{\mathfrak{p} \in S} |m_{\mathfrak{p}}(t)| \leq \exp\left(-\frac{1}{8}g(t)\right), \quad g(t) := \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} |1 - N\mathfrak{p}^{i\beta t}|^2.$$

Now ψ is the characteristic function of the law of X under ν , so it is continuous with $\psi(0) = 1$; choose $\delta > 0$ with $|\psi| \geq \frac{1}{2}$ on $[0, \delta]$. Then $g \leq 8 \log 2$ on $[0, \delta]$. But $|1 - e^{i\omega t}|^2 = 4 \sin^2(\omega t/2)$ and $\int_0^\delta 4 \sin^2(\omega t/2) dt = 2\delta - \frac{2}{\omega} \sin(\omega\delta) \geq \delta$ as soon as $\omega\delta \geq 2$, so by Tonelli

$$\int_0^\delta g(t) dt \geq \delta \sum_{\mathfrak{p} \in S: \beta\delta \log N\mathfrak{p} \geq 2} N\mathfrak{p}^{-\beta} = \infty,$$

the sum omitting only finitely many primes of S and $\zeta_{K,S}(\beta) = \infty$. This contradicts $g \leq 8 \log 2$ on $[0, \delta]$, so no such f exists and M is of type III.

For the last statement, a closed subgroup of $\lambda_0^{-\beta\mathbb{Z}}$ is $\lambda_0^{-n\beta\mathbb{Z}}$ for some $n \geq 0$; $n = 0$ would give $S(M) \subseteq \{0, 1\}$, i.e. type III₀, and $n \geq 1$ gives type III _{$\lambda_0^{-n\beta}$} . $\square$

Remark 6.2. Proposition 6.1 bounds $S(M)$ from *above*. It is tempting to argue that when two norms have irrational log-ratio, Γ_β is dense in $\mathbb{R}_{>0}$ and therefore $S(M) = [0, \infty)$, i.e. M is III₁; but this inference is invalid, since density of the group of *possible* ratios gives no lower bound on the ratio set, and III₀—where $S(M) = \{0, 1\}$ —is not excluded. Establishing III₁ requires the ergodicity of the Maharam extension of the action, equivalently that the essential range of the cocycle is all of $\overline{\Gamma_\beta}$.

The natural candidate for the missing ingredient is an *extended* obstruction group $\Xi_\beta^{\text{ext}} \leq \widehat{G_S} \times \mathbb{R}$, obtained from (1.1) by twisting the characters with $N\mathfrak{p}^{i\beta s}$ and having Ξ_β as its $s = 0$ slice; the essential range of the cocycle should be readable from its projection to $\mathbb{R}$. We do not pursue this here, and nothing in this paper depends on it. Note that the commensurable case of Proposition 6.1 is vacuous once S is infinite, since $\log p$ for distinct rational primes are $\mathbb{Q}$ -linearly independent and only finitely many primes of K lie above each p ; any failure of type III₁ arises from *approximate* commensurability along a set of divergent β -weight.

Open problems.

- (1) Determine the type of the extremal states. Proposition 6.1 bounds $S(M)$ from above and Remark 6.2 explains what a lower bound requires; for the sparse sets of §5 it is not decided here which of the alternatives III₁, III _{λ} , III₀ occurs.
- (2) Characterize the possible chains $\{\Xi_\beta\}_{\beta > 0}$. By Lemma 3.5 they are increasing; Proposition 4.16 realizes finite chains. The transition set is always countable, $\widehat{G_S}$ being countable; which monotone families of subgroups of $\widehat{G_S}$ occur, and whether infinitely many transitions can be forced into a bounded interval, are open.
- (3) Extend the criterion to the function field case, where Chebotarev is available with error terms from the Riemann hypothesis for curves, and where sparse sets of closed points can be constructed with more freedom than sparse sets of primes. One structural difference is worth recording: the log-norms $\log NP = \deg P \cdot \log q$ are pairwise commensurable, so the commensurable case of Proposition 6.1—vacuous over a number field by Remark 6.2—is there the only case.

- (4) In Example 4.2 the obstruction is unramified. Is there a canonical description of the sub-simplex of KMS_β states cut out by the unramified part of $\widehat{G}_S$, in terms of the Hilbert class field, valid for all S simultaneously?

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GUT GEOSERVICE INC., MONTREAL, CANADA
 Email address: ruqing@hotmail.com